Keep Flying Whatever the Model Says: Predictive Safety Filtering for Language-Model UAV Agents under Adaptive Attack

Anonymous submission
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***Abstract*—Large language model (LLM) agents are entering the planning loop of uncrewed aerial vehicles (UAVs), where any text they read (an operator note, a map annotation, a sign in a camera frame) is an attack surface. Adaptive attacks keep breaking model-level defences against prompt injection, so we stop asking whether the model can be secured and make physical safety independent of it. A predictive safety filter (PSF) sits between the agent and the autopilot. Every period it solves a small quadratic program (QP): it executes the agent's proposal when a model-predictive trajectory that begins with it keeps the *true* position, not the estimate, inside a tightened keep-in set and ends in a robustly invariant loiter set. Otherwise it applies the closest admissible input, and if no verified plan is found it continues a latched certified plan. Theorem 1 proves safety of the true state for *every* sequence of planner outputs, under a stated model-mismatch bound, a sound protection level with a staleness look-ahead, and a covert sensing budget. Safety is therefore unconditional within these assumptions; liveness is not. Proposition 2 splits the residual mission denial into a filter-induced part and an in-envelope part that no physical filter can remove. Proposition 3 proves that the admissible sets are nested under the sensing budget. We also give an eight-level attacker ladder that ends in an action-space oracle; by Theorem 1 this oracle upper-bounds every prompt-injection attacker for the safety claim. In a simulation pilot with non-LLM planners, the filter had zero true-state violations in 3,608,500 aircraft-steps (one-sided 95% Clopper–Pearson bound 8.3×10^{-7}), while oracle attackers drove the aircraft to within 0.062 m of the boundary. Without the filter, every oracle episode violated. The benign intervention rate was 7.7%, at 0.14 s of mission time. A planner that proposes only safe holds still denied every mission while the filter stayed transparent. Violations reappeared exactly when the mismatch bound was exceeded. Campaigns with real language models are specified but have not yet been run.**

***Index Terms*—LLM agents, prompt injection, safety filters, model predictive control, runtime assurance, UAV security, adaptive attacks.**

I. Introduction

Language-model agents are moving from chat windows into the control hierarchy of robots and aircraft. Large language models (LLMs) now write programs for small drones [1], [2], select skills for manipulators [3], [4], and, in one aerial system, write an additive term directly into a multirotor's linear-quadratic regulator at 0.1–1 Hz [5]. Surveys of LLMs for uncrewed aerial vehicles (UAVs) treat this as the next step in autonomy [6]. Every one of these systems reads text that an adversary can influence: an operator message, a map annotation, a peer report, or a sign in the camera's field of view.

That text channel is demonstrably exploitable. Indirect prompt injection [7] lets anyone who controls data the agent ingests steer what it does. Jailbreaks against LLM-controlled robots succeed with high probability [8]–[10]. Command Hijacking against Embodied AI (CHAI) [11] hijacks a simulated drone's emergency landing in up to 72.8% of trials and aerial tracking in up to 95.5%, using printed signs alone. CHAI evaluates no defence and calls for provably robust ones. Alignment is not that defence. In a drone-control benchmark, GPT-3.5 refused only 12.50% of harmful requests, and even 13–34B models refused only about 42% of unintentional hazards [12].

Model-level defences against prompt injection have a poor record under adaptive attack. Nasr, Carlini, Tramèr and colleagues broke twelve recent jailbreak and prompt-injection defences, most of them at attack success rates (ASRs) above 90%, many of which had originally reported near-zero ASR. Their rule is blunt: the attacker moves second [13]. Zhan et al. bypassed all eight indirect injection defences they tested [14]. A learned attacker reaches 98% ASR against GPT-4o and 72% against GPT-5 with an instruction hierarchy [15]. The only robot guardrail evaluated against adaptive attackers, RoboGuard [16], is semantic. It checks plans against linear temporal logic (LTL) specifications that a root-of-trust LLM writes. Its authors note that "LTL may not capture safety tied to system dynamics". Physical filters for LLM-driven robots exist [17]–[20], but none of them has been evaluated against an adaptive attacker, and none states a guarantee quantified over an *adversarial* planner.

a) The move: We stop asking whether the language model can be kept from being fooled. Instead we make the aircraft's physical safety a property that does not depend on the answer. We borrow the logic of Simplex [21], [22] and of predictive safety filters [23], [24]. The agent is an untrusted high-level planner. Between its proposals and the autopilot we place a filter that trusts only a small, analysable base: the plant model, the navigation certificate, a convex solver, an exact verifier and a loiter fallback. The agent may propose anything. The filter executes a proposal only when a certified trajectory beginning with it exists. Otherwise it executes the closest input that has one. If its solver fails, it continues a plan it has already certified. Whether an attacker fooled the model then says nothing about whether the aircraft stays in its envelope. It speaks only to whether the mission finishes.

This is a security paper about a control object. We realise the runtime-assurance direction that the programme's parent manuscript [25] names for its persistent-denial gap: a fallback

"whose guarantee is independent of" the navigation Lipschitz constant. The filter is a physical-layer member of the family of "defence by design" system-level defences [26]–[28], which bound what a compromised model can do rather than trusting it to resist injection.

b) Contributions:

C1 Planner-invariant robust safety of the true state (Theorem 1). A predictive safety filter (PSF) for a UAV fleet whose guarantee holds for every planner sequence, adaptive and filter-aware. The guarantee carries an explicit model-mismatch margin, the programme's time-varying protection level $\rho_i(t)$ [29] with a staleness look-ahead, a covert sensing budget [30], a continuation problem that restores recursive feasibility under re-anchoring, and an independently verified, latched fallback.

C2 An honest account of what the attacker keeps (Propositions 2 and 3). A safety/liveness decomposition in the sense of Alpern and Schneider [31], with a denial taxonomy that separates filter-induced denial from in-envelope denial. We show that no filter that is transparent to its own fallback can remove the latter. We also prove that the admissible sets are nested in the sensing budget. Monotonicity of the closed-loop denial rate is stated as a hypothesis and tested, not claimed.

C3 An adaptive evaluation protocol for physical-layer defences (Section IV). An eight-level attacker ladder (A0–A7), from a static injection corpus to an action-space oracle (A6) and its composition with a covert sensing attacker (A7). The oracle is a free upper envelope for the safety claim (Corollary 4) and turns "we cannot prove we found the strongest attack" into a measurable *mediation slack*.

C4 A pilot with real numbers and a pre-registered LLM campaign (Sections V and VI). A seeded simulation pilot with non-LLM planners (1,489 episodes) exercises every part of the theory, including the negative controls that break each assumption. The real-LLM campaign is specified, budgeted and marked **[pending]** wherever its results do not exist.

c) Summary of findings (pilot): Inside the stated assumptions the filter recorded zero violations in 3,608,500 aircraft-steps. Oracle attackers pushed the true state to within 0.062 m of the keep-in boundary and 0.383 m of the separation minimum, so the tightening is nearly tight and the attack is not weak. The same planners violated in 40/40 episodes without the filter. Safety, however, is the easy half. A planner that proposes only safe holds denies 100% of missions while the filter stays essentially transparent. A filter-aware oracle that keeps a progress component converts its blocked safety attack into filter-induced denial in 95% of four-aircraft missions. Violations reappear exactly when an assumption is broken: a mismatch at twice the declared bound, certificate jumps, or a look-ahead removed by ablation. The headline is therefore a pair. *Safety is unconditional under the stated model; liveness is not.*

II. Preliminaries and Problem Statement

A. Notation and programme context

For $x \in \mathbb{R}^n$, $\|x\|_\infty$ and $\|x\|_1$ are the usual norms, and $|x|$ is taken elementwise. Inequalities between vectors hold componentwise. A fleet $\mathcal{N} = \{1, \dots, N\}$ flies in the plane. The programme supplies three quantities that this paper consumes but does not re-derive:

- The parent manuscript's Lipschitz–Grönwall chain [25], with delay-to-position rate $\gamma = 1.625$ m/s (the supremum over 674 post-onset samples of real jamming and spoofing flights), local Lipschitz constant $\hat{L}_{\mathrm{loc}} = 1.1814$ and certification horizon $T_{\mathrm{c}} = 1$ s, so that $e^{\hat{L}_{\mathrm{loc}} T_{\mathrm{c}}} = 3.259$.
- Paper III's protection level $\rho_i(t)$ [29].
- Paper IV's perturbation budget [30].

We write κ_{sched} for the parent's schedule tolerance, to avoid a clash with the four other uses of κ in the programme. We write N_{p} for the filter horizon; H is reserved for malicious relay hops. A full notation table is in Section C and an acronym list in Section D.

B. Plant with bounded model mismatch

Each aircraft i carries a navigation state $\hat{x}_i = (\hat{p}_i, \hat{v}_i) \in \mathbb{R}^2 \times \mathbb{R}^2$, which the control loop closes on. The low-level loop runs at step δt; time $t_n = n\,\delta t$. Per axis, the plant as seen by the navigation solution is

$$\hat{x}_i(n{+}1) = A_\delta \hat{x}_i(n) + B_\delta\big(u_i(n) + d_i(n)\big), \quad \|d_i(n)\|_\infty \le \bar{w}, \tag{1}$$

with $A_\delta = \left[\begin{smallmatrix} 1 & \delta t \\ 0 & 1 \end{smallmatrix}\right]$ and $B_\delta = \left[\begin{smallmatrix} \delta t^2/2 \\ \delta t \end{smallmatrix}\right]$ applied to each axis. The actuator command u_i (an acceleration) saturates at $u_{\max}$ per axis. Unlike the parent, where u is a sensor input, u here is the actuator command. The disturbance d_i lumps all model–plant mismatch: drag, wind and unmodelled dynamics. Its bound $\bar{w}$ is the *model-mismatch margin* that the filter must carry explicitly.

The *true* position differs from the navigation estimate:

$$p_i(n) = \hat{p}_i(n) + e_i^{\mathrm{cert}}(n) + e_i^{\mathrm{adv}}(n). \tag{2}$$

Here e_i^{cert} is the estimation error that the certificate bounds, and e_i^{adv} is a covert sensing offset (Section II-E). Safety is judged on p_i, never on $\hat{p}_i$.

C. Safe set

The keep-in set is a polytope $\mathcal{C} = \{p \in \mathbb{R}^2 : h_r^\top p \le c_r,\ r = 1, \dots, n_h\}$, with unit outward normals h_r. The programme's corridor of half-width m [25] is the two-half-plane special case. Pairs must keep a true separation of at least d_{sep}. The fleet's safe set at step n is

$$\mathcal{S} = \big\{(p_1, \dots, p_N) : p_i \in \mathcal{C}\ \forall i,\ \|p_i - p_l\|_2 \ge d_{\mathrm{sep}}\ \forall i \ne l\big\}. \tag{3}$$

D. Navigation certificate with staleness

Paper III [29] reports, for every aircraft, a protection level

$$\rho_i(t) = e^{\hat{L}_{\mathrm{loc}} T_{\mathrm{c}}}\, \gamma\, \Delta_i(t), \qquad \|\hat{p}_i(t) - p_i^{\mathrm{nav\,true}}(t)\|_2 \le \rho_i(t), \tag{4}$$

where $\Delta_i(t)$ is the staleness, the age of the newest trustworthy fix. Under jamming it grows at 1 s/s, so ρ_i grows at most at the slope $L_\rho := e^{\hat{L}_{\mathrm{loc}} T_{\mathrm{c}}}\gamma = 5.296$ m/s. A trustworthy fix re-anchors it to a benign floor $\rho_{\mathrm{fl}} = L_\rho \cdot 0.178\,\mathrm{s} = 0.943$ m, where 0.178 s is the largest benign residual in the parent's hop ledger. Paper III treats $\rho_i(\cdot)$ as an adversarially chosen càdlàg process, and notes that "jumps cannot be predicted from a rate bound": the median undetected two-hop relay jump is about 41.8 m. We declare a staleness envelope $\bar{\Delta}$, so that $\rho_i \le \bar{\rho} := L_\rho \bar{\Delta}$. The filter tightens by box errors, and the box is conservative with respect to the Euclidean certificate because $\|e\|_\infty \le \|e\|_2$.

E. Sensing budget

Paper IV composes its perturbation sources into a budget $\delta_{\mathrm{tot}} = \delta_{\mathrm{net}}(\theta) \oplus \delta_{\mathrm{adv}}(\theta) \oplus \delta_{\mathrm{gnss}} \oplus \delta_{\mathrm{fault}} \oplus \delta_{\mathrm{env}}$ whose $\oplus$ has mixed semantics [30]. Deterministic terms shift the margin. Stochastic and probability-box (p-box) terms enter through a dependence-sharp tail bound $\bar{p}$. Paper IV never reduces δ_{tot} to a scalar. Following its own design rule, we define the scalar tightening

$$b_{\mathrm{tot}}(\varepsilon) = e^{\hat{L}_{\mathrm{loc}} T_{\mathrm{c}}} \delta_{\mathrm{net}}(\theta) + \delta_{\mathrm{adv}}(\theta) + s^\star(\varepsilon), \tag{5}$$

where $s^\star(\varepsilon) = \inf\{s : \bar{p}(s) \le \varepsilon\}$. At $\theta = 0.25$ s and $H = 2$ hops, $e^{\hat{L}_{\mathrm{loc}} T_{\mathrm{c}}} \delta_{\mathrm{net}} = 2.648$ m. The dependence-sharp union bound over Paper IV's committed marginals gives $s^\star(0.1) = 5.70$ m. The stealth budget δ_{adv} is empirical. The pilot used Paper IV's first published value, $\delta_{\mathrm{adv}} = 4.371$ m. Paper IV's later re-pooling of its attacker search gives $\delta_{\mathrm{adv}} = 5.463$ m at a 1% or 5% false-positive rate (FPR) and 5.038 m at 10%. We treat 5.463 m as the headline for the planned campaign and 4.371 m as a sensitivity level (Section VI-D). Paper IV certifies *covert* attackers only, that is, on the event of no alarm and under a trusted re-anchoring fix (its condition C2′). Accordingly, the covert offset of Eq. (2) satisfies $\|e_i^{\mathrm{adv}}\|_\infty \le b_{\mathrm{adv}} \le b_{\mathrm{tot}}$.

F. The language-model planner as an untrusted map

Every planning period the agent receives a situation description made of delimited textual channels: trusted telemetry, an operator note, map annotations and peer reports. It returns one structured action per aircraft from a JavaScript Object Notation (JSON) schema: `goto` (waypoint, speed), `velocity`, or `hold`. Malformed output becomes `hold`. We model the agent as an arbitrary, possibly randomised map from the full interaction history to actions. A fixed nominal tracking map π_{nom} converts an action into a desired acceleration, $u_i^{\mathrm{des}} = \pi_{\mathrm{nom}}(\hat{x}_i, a_i)$. Nothing else from the agent reaches the filter.

G. Threat model

a) Capabilities: The adversary controls any subset of the untrusted text channels. In the strongest case (A6) it chooses the action stream directly. It knows the filter, the plant model, the constraint set, the solver, the certificate and the current state (Kerckhoffs). Its choices may depend on everything it has observed, including whether the filter intervened. In the composed case (A7) it also adds a covert sensing offset within Paper IV's budget.

b) Goals, in order of severity: (G1) a true-state safety violation; (G2) maximal fallback; (G3) mission denial that an operator-side monitor does not attribute to an attack (stealthy denial); (G4) boundary riding, that is, keeping the fleet at the edge of the admissible set.

c) Trust boundary: The LLM, its prompts and every text channel are untrusted. The trusted computing base (TCB) is the plant model with its mismatch bound, the certificate process, the filter's QP and verifier, the tube controller and the fallback. All are small enough to analyse; the LLM is not. We secure the *system*, not the model.

d) Out of scope: The following are stated as assumptions, not defended: attacks on the TCB itself; certificate jumps beyond the declared envelope; sensing attackers that are indifferent to detection (Paper IV's covert-only scope); and estimator attacks that move the navigation state beyond the mismatch bound. The last class is exactly the "stealthy deactivation" of safety filters [32].

H. Problem statement

Let σ denote a closed-loop trace. The set $S = \{\sigma : p(n) \in \mathcal{S}\ \forall n\}$ is a *safety* property: every violating trace has a finite bad prefix [31], [33]. Let $L = \{\sigma :$ every aircraft eventually reaches its goal$\}$; this is a *liveness* property, since every finite prefix can be extended into L. Missions carry a deadline $(1 + \kappa_{\mathrm{sched}})T$. Following the parent, the mission completion rate is conjunctive,

$$\mathrm{MCR} = \Pr\big[\sigma \in S \ \wedge\ t_{\mathrm{arr},i} \le (1 + \kappa_{\mathrm{sched}})T\ \forall i\big], \tag{6}$$

and the mission denial rate is $\mathrm{MDR} = 1 - \mathrm{MCR}$. A deadline makes completion a bounded requirement, which is technically a safety property in the Alpern–Schneider topology; it is not, however, one the filter enforces. We seek a filter and an evaluation that answer:

- **RQ1** Does $\sigma \in S$ hold for every planner, empirically and under the strongest adaptive attacker we can build?
- **RQ2** Which denial can an adaptive attacker force, through which mechanism, and what does the filter do about each?
- **RQ3** How do the admissible sets and the induced denial change as a covert sensing budget composes with the language attack?
- **RQ4** What does the filter cost when there is no attack?

III. The Predictive Safety Filter and Its Guarantees

A. Two time scales and the tube controller

The filter runs every $T_{\mathrm{s}} = S\,\delta t$ (in the pilot $\delta t = 0.1$ s, $T_{\mathrm{s}} = 0.5$ s, $S = 5$). It plans a nominal trajectory $z_{i,k} =$

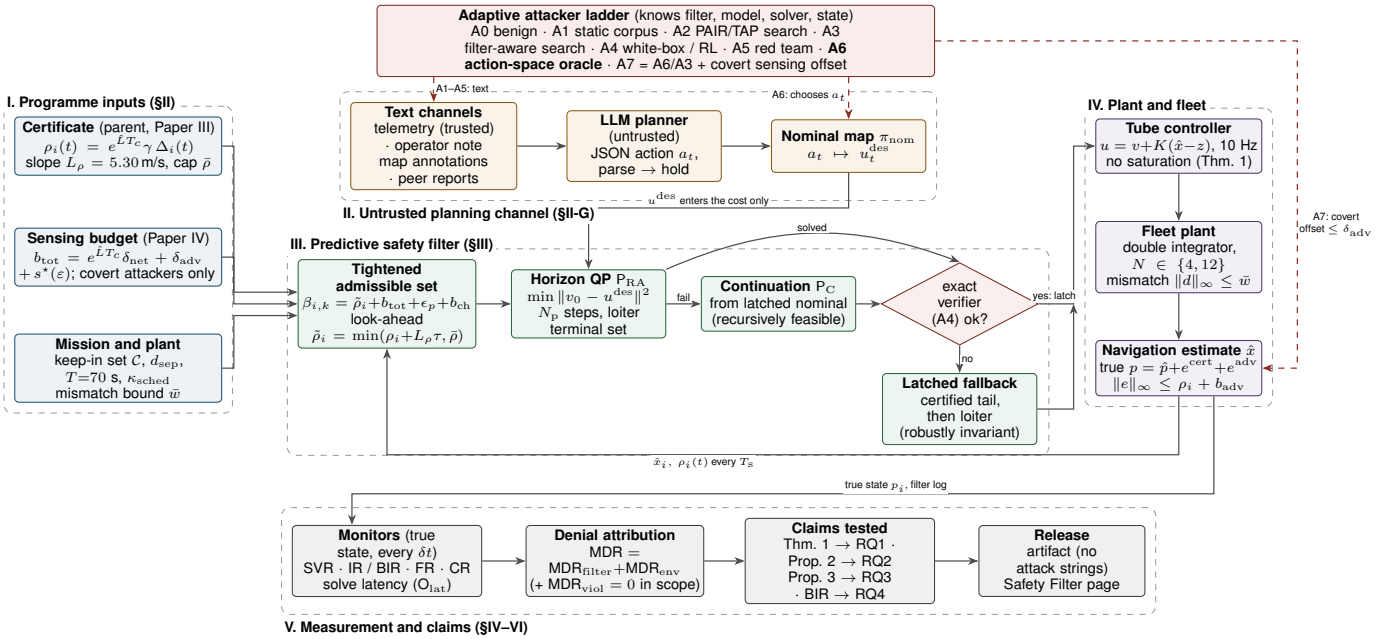


Fig. 1. Methodological framework. **I.** The programme supplies the navigation certificate $\rho_i(t)$ with its staleness slope L_ρ, the covert sensing budget $b_{\rm tot}$ and the mission; the plant's mismatch bound $\bar w$ is new. **II.** The untrusted LLM planner reads delimited text channels and emits a JSON action that a fixed nominal map turns into a desired acceleration $u^{\rm des}$. Attackers A1–A5 act through text; the oracle A6 chooses the action directly; A7 adds a covert sensing offset (dashed red). **III.** The filter tightens the keep-in and separation constraints by the look-ahead protection level, the sensing budget, the mismatch tube and the inter-node chord, then solves a horizon QP in which $u^{\rm des}$ enters only the cost. If the re-anchored problem fails, a continuation problem from the latched nominal is solved. Only plans that an exact verifier accepts are latched; otherwise the fleet continues the last certified plan and then loiters. **IV.** A tube controller tracks the latched nominal. **V.** Monitors evaluate the *true* state at every plant step and attribute every denied mission to the filter or to the planner; the measurements test Theorem 1 and Propositions 2–3 and answer RQ1–RQ4.

$(p(z_{i,k}), \mathrm{vel}(z_{i,k}))$ over $N_{\rm p}$ intervals with piecewise-constant nominal accelerations $v_{i,k}$:

$$z_{i,k+1} = A_{\rm s} z_{i,k} + B_{\rm s} v_{i,k}, \quad A_{\rm s} = \left[\begin{smallmatrix} 1 & T_{\rm s} \\ 0 & 1 \end{smallmatrix}\right], \; B_{\rm s} = \left[\begin{smallmatrix} T_{\rm s}^2/2 \\ T_{\rm s} \end{smallmatrix}\right]. \tag{7}$$

Between ticks, a tube controller tracks the latched nominal $z(n)$ on the fine grid,

$$u_i(n) = v_i(n) + K\big(\hat x_i(n) - z_i(n)\big), \tag{8}$$

with $K = -[k_p \;\; k_d] = -[1.2 \;\; 1.8]$. The gains are Paper IV's lateral PD gains [30]. The tracking error $\xi_i = \hat x_i - z_i$ obeys $\xi_i(n{+}1) = \Phi\,\xi_i(n) + B_\delta d_i(n)$ with $\Phi = A_\delta + B_\delta K$, whose spectral radius is $\lambda_c = 0.909$. After n steps from an anchor where $\xi = 0$, the per-axis error is bounded by the reachable-set sums

$$\epsilon_\star(n) = \bar w \textstyle\sum_{q=0}^{n-1} |c_\star \Phi^q B_\delta|, \quad c_p = e_1^\top, \; c_v = e_2^\top, \; c_u = K, \tag{9}$$

which converge because Φ is Schur. This is a Chisci-type tightening [34] for a tube controller [35]; like the contraction-based tightening of Köhler et al. [36], it stays bounded in the horizon. The Lipschitz-based alternative [37] grows geometrically. At $\bar w = 0.5\,\mathrm{m/s^2}$: $\epsilon_p(\infty) = 0.424\,\mathrm{m}$, $\epsilon_v(\infty) = 0.400\,\mathrm{m/s}$ and $\epsilon_u(\infty) = 0.690\,\mathrm{m/s^2}$. The nominal acceleration is therefore limited to $u_{\rm nom} = u_{\max} - \epsilon_u(\infty) = 3.310\,\mathrm{m/s^2}$, which guarantees that Eq. (8) never saturates.

B. Constraint tightening

Consider a plan created at tick t_j whose tube was last anchored a fine steps earlier ($a = 0$ for a plan that starts at the navigation state). Using Paper III's look-ahead [29], the predicted protection level is

$$\tilde\rho_i(t_j;\tau) = \min\big\{\max\{\rho_i(t_j), \rho_{\rm fl}\} + L_\rho\tau, \; \bar\rho\big\}. \tag{10}$$

It covers growth at slope L_ρ, a re-anchor to at most $\rho_{\rm fl}$ within the horizon, and the cap $\bar\rho$. The per-axis tightening at node k is

$$\beta_{i,k} = \begin{cases} \tilde\rho_{i,k} + b_{\rm tot} + \epsilon_{p,k} + b_{\rm ch}, & k < N_{\rm p}, \\ \bar\rho + b_{\rm tot} + \epsilon_p(\infty) + b_{\rm ch}, & k = N_{\rm p}, \end{cases} \tag{11}$$

with $\tilde\rho_{i,k} = \tilde\rho_i(t_j; (k{+}1)T_{\rm s})$ and $\epsilon_{p,k} = \epsilon_p(a + (k{+}1)S)$, and $b_{\rm ch} = u_{\rm nom} T_{\rm s}^2/8$ bounds the deviation of the nominal path from the chord between two nodes. Every term is non-decreasing in k, so $\beta_{i,k}$ covers the whole interval that follows node k. Node constraints use $\hat\beta_{i,0} = \beta_{i,0}$ and $\hat\beta_{i,k} = \max\{\beta_{i,k-1}, \beta_{i,k}\}$, so that both ends of every interval are tightened for that interval. Speeds are tightened by ϵ_v and inputs by $\epsilon_u(\infty)$. At $b_{\rm tot} = 4.371\,\mathrm{m}$ the terminal tightening is $9.898\,\mathrm{m}$.

C. The filter problem

At tick t_j, with start state ζ_i and tube age a, the filter solves

$$\min_{\{v_{i,k}\}} \ \sum_i \|v_{i,0} - u_i^{\mathrm{des}}\|_2^2 + \mu \sum_i \sum_{k=1}^{N_\mathrm{p}-1} \|v_{i,k}\|_2^2 \tag{12a}$$

$$\text{s.t. } z_{i,0} = \zeta_i, \quad z_{i,k+1} = A_\mathrm{s} z_{i,k} + B_\mathrm{s} v_{i,k}, \tag{12b}$$

$$h_r^\top p(z_{i,k}) \le c_r - \|h_r\|_1 \hat{\beta}_{i,k} \quad \forall r,\ k = 0, \dots, N_\mathrm{p}, \tag{12c}$$

$$n_{il,k}^\top \big(p(z_{i,q}) - p(z_{l,q})\big) \ge d_\mathrm{sep} + \|n_{il,k}\|_1 \big(\bar{\beta}_{i,kq} + \bar{\beta}_{l,kq}\big),\ q \in \{k, k{+}1\}, \tag{12d}$$

$$|v_{i,k}| \le u_\mathrm{nom}, \quad |\mathrm{vel}(z_{i,k})| \le v_\mathrm{max} - \epsilon_{v,k}^+, \tag{12e}$$

$$\mathrm{vel}(z_{i,N_\mathrm{p}}) = 0 \qquad \text{(loiter terminal set)}, \tag{12f}$$

for all $i < l$ and $k < N_\mathrm{p}$ in (12d), with $\bar{\beta}_{i,kq} = \max\{\beta_{i,k}, \beta_{i,q}\}$, $\epsilon_{v,k}^+ = \epsilon_v(a + \min\{k{+}1, N_\mathrm{p}\}S)$ and $\mu = 10^{-4}$. The separation constraint is a linearisation along a unit normal $n_{il,k}$ taken from a reference trajectory. For any unit normal it implies $\|p_i - p_l\|_2 \ge n^\top (p_i - p_l)$, so it is conservative. Constraint (12d) is imposed at both ends of every interval with the same normal, so it holds along the whole relative chord. The planner's proposal enters (12) *only* through u^{des} in the cost. The feasible set does not depend on it, and this is the structural fact behind the quantifier "for every planner".

a) Re-anchoring, continuation and fallback: Each tick runs up to three stages (Algorithm 1).

1) **Re-anchored problem** P_RA: start at the navigation state, $\zeta_i = \hat{x}_i(t_j)$, with tube age $a = 0$, and normals from the latched plan shifted to the current state.
2) **Continuation problem** P_C: if P_RA fails, start from the latched nominal state, $\zeta_i = z_i^{\mathrm{lat}}(t_j)$, with the accumulated tube age and the latched plan's own normals, shifted by one tick.
3) **Fallback**: if both fail, execute the remaining tail of the latched certified plan, then loiter at its terminal node.

Only stage 3 counts as a fallback. Every candidate plan is re-simulated and checked by an independent evaluator that uses exact arithmetic and none of the solver's numerical buffers. A plan that fails this check is treated as a solver failure. Re-anchoring alone does not preserve recursive feasibility under worst-case disturbance near the boundary: the new tightening counts error that has already been realised. The continuation problem is the remedy, in the spirit of tube MPC with a free initial nominal state [35]. Black-box Simplex [38] supplies the idea of keeping the last valid certificate.

D. Theorem 1: planner-invariant robust safety

Assumption A1 (Bounded mismatch)**.** $\|d_i(n)\|_\infty \le \bar{w}$ for all i, n in (1).

Assumption A2 (Sound, rate-bounded certificate; covert budget)**.** $\|e_i^{\mathrm{cert}}(n)\|_\infty \le \rho_i(t_n)$. Between re-anchors, ρ_i grows at most at slope L_ρ; a re-anchor resets it to at most ρ_fl; and $\rho_i \le \bar{\rho}$ always, with no jumps ($\bar{J} = 0$, the delay-adversary regime). The sensing offset satisfies $\|e_i^{\mathrm{adv}}\|_\infty \le b_\mathrm{adv} \le b_\mathrm{tot}$ (Paper IV's covert class).

Algorithm 1 One filter tick at t_j (all aircraft jointly)

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u^des ← π_nom(x̂, a_t)                    ▷ untrusted input, cost only
(ok, V) ← solve P_RA from x̂(t_j), age 0; verify exactly
if not ok and a certified latched plan exists then
    (ok, V) ← solve P_C from z^lat(t_j), age a^lat; verify
end if
if ok then latch (z, V)                  ▷ certified, executed by (8)
else keep latched plan; its tail, then loiter      ▷ fallback
end if
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Assumption A3 (Well-posed tightening)**.** Φ is Schur, $\epsilon_u(\infty) < u_\mathrm{max}$ and $\epsilon_v(\infty) < v_\mathrm{max}$.

Assumption A4 (Verified execution)**.** A plan is latched only after an independent evaluator has checked every constraint of (12) on its rollout. Solver failure, time-out or a failed check is reported as failure and never executes.

Assumption A5 (Initial certification)**.** At t_0 the problem P_RA has a verified solution.

Theorem 1 (Planner-invariant robust safety)**.** *Assumptions A1 to A5 hold. Consider every planner: any, possibly randomised, map from the full history (including the filter's internal state, the solver, the certificate, the disturbance and the sensing offset) to action sequences. Consider also every disturbance allowed by Assumption A1 and every certificate and offset trajectory allowed by Assumption A2. Then, at every plant step n and for all $i \ne l$:*

(i) the true position satisfies $p_i(n) \in \mathcal{C}$;

(ii) $\|p_i(n) - p_l(n)\|_2 \ge d_\mathrm{sep}$;

(iii) $|u_i(n)| \le u_\mathrm{max}$ and $|\hat{v}_i(n)| \le v_\mathrm{max}$, so the actuators never saturate;

(iv) whenever a certified plan is latched at t_j, P_C is feasible at t_{j+1}, so a fallback can only follow a solver or verifier failure, never infeasibility; a fallback preserves (i)–(iii) indefinitely.

The proof rests on three lemmas. Full index bookkeeping is in Section A.

Lemma 1 (Tube)**.** *If $\xi_i = 0$ at an anchor and (8) tracks a nominal trajectory that is continuous across ticks, then n steps later $|e_1^\top \xi_i| \le \epsilon_p(n)$, $|e_2^\top \xi_i| \le \epsilon_v(n)$ and $|K\xi_i| \le \epsilon_u(n)$ per axis, provided no saturation occurs.*

Proof. Without saturation, $\xi_i(n) = \sum_{q=0}^{n-1} \Phi^q B_\delta d_i(n{-}1{-}q)$. Apply the triangle inequality with $|d_i| \le \bar{w}$ per axis. The dynamics decouple per axis. □

Lemma 2 (A certified plan is safe while followed)**.** *Let a plan be verified at t_j from start ζ with tube age a, where $|\xi(t_j)|$ is bounded by the tube at age a. Suppose the plan is followed by (8) and then held at its terminal node with $v = 0$ for any length of time. Then (i)–(iii) hold at every step of that execution.*

Proof. Take a step n in interval k, so $t_j + kT_\mathrm{s} \le t_n \le t_j + (k{+}1)T_\mathrm{s}$. By (2), $p_i(n) = z_i(n) + e_1^\top \xi_i(n) + e_i^{\mathrm{cert}}(n) + e_i^{\mathrm{adv}}(n)$. Each axis of $z_i(n)$ lies within b_{ch} of the chord point between $p(z_{i,k})$ and $p(z_{i,k+1})$, because the nominal acceleration is bounded by u_{nom} (Assumption A4 enforces (12e)). By Lemma 1 and monotonicity, $|e_1^\top \xi_i(n)| \le \epsilon_p(a + (k{+}1)S)$.

By Assumption A2, $\rho_i(t_n) \le \tilde{\rho}_i(t_j; (k{+}1)T_\mathrm{s})$. If no re-anchor occurs in $[t_j, t_n]$, ρ_i grows by at most $L_\rho(t_n - t_j)$. If one occurs at time s, then $\rho_i(t_n) \le \rho_{\mathrm{fl}} + L_\rho(t_n - s)$. In both cases the cap is $\bar{\rho}$. Hence every axis of $p_i(n)$ lies within $\beta_{i,k}$ of the chord point.

Both chord endpoints satisfy (12c) with $\hat{\beta} \ge \beta_{i,k}$, and the half-space $\{y : h_r^\top y \le c_r - \|h_r\|_1 \beta_{i,k}\}$ is convex. The chord point therefore satisfies it, and $h_r^\top p_i(n) \le h_r^\top (\text{chord}) + \|h_r\|_1 \beta_{i,k} \le c_r$: claim (i). Separation follows in the same way. The constraint (12d) holds at both ends of the interval with the same $n_{il,k}$ and with $\bar{\beta} \ge \beta_{\cdot,k}$, so it holds on the relative chord. The relative deviation is at most $\|n\|_1(\beta_{i,k} + \beta_{l,k})$ along n, and $\|p_i - p_l\|_2 \ge n^\top (p_i - p_l) \ge d_{\mathrm{sep}}$: claim (ii).

For claim (iii), the nominal velocity is affine within an interval, so it is bounded by its node values; together with $|e_2^\top \xi| \le \epsilon_v$ this gives $|\hat{v}_i| \le v_{\max}$. Also $|u| \le |v| + |K\xi| \le u_{\mathrm{nom}} + \epsilon_u(\infty) = u_{\max}$, which also justifies the no-saturation premise of Lemma 1 by induction over steps.

After the horizon, the nominal rests at z_{i,N_p}, $|\xi| \le \epsilon(\infty)$ and $\rho_i \le \bar{\rho}$. These are covered by β_{i,N_p} and by the terminal-node constraints: the loiter set is robustly invariant. □

Lemma 3 (Shift feasibility of the continuation problem)**.** *Suppose a plan (z, v) with normals $n_{\cdot,k}$ is certified at t_j with age a. At t_{j+1}, take the shifted candidate $z'_k = z_{k+1}$ and $z'_{N_\mathrm{p}} = z_{N_\mathrm{p}}$, with $v'_k = v_{k+1}$ for $k < N_\mathrm{p} - 1$ and $v'_{N_\mathrm{p}-1} = 0$, and normals $n'_k = n_{\min\{k+1, N_\mathrm{p}-1\}}$, at age $a + S$. This candidate is feasible for* P_C.

The proof is index bookkeeping: tube terms match by an index shift, Assumption A2 makes the shifted look-ahead no larger, and the appended node is the old terminal node at rest (Section A).

Proof of Theorem 1. We argue by induction over ticks. The invariant $\mathcal{I}_j$ is: *at t_j a certified plan is latched, and its continued execution followed by loitering satisfies (i)–(iii).* $\mathcal{I}_0$ holds by Assumption A5 and Lemma 2.

Assume $\mathcal{I}_j$. At t_{j+1} exactly one of three events occurs:

- P_{RA} returns a verified plan. Its start is the navigation state, so $\xi = 0$ and the tube age is 0; Lemma 2 applies.
- P_C returns a verified plan. Its start is the latched nominal state, which is continuous with the executed nominal, so $\xi(t_{j+1})$ is bounded by the tube at the accumulated age; Lemma 2 applies again.
- Neither returns a verified plan. The old plan stays latched and $\mathcal{I}_j$ carries over unchanged.

In every case $\mathcal{I}_{j+1}$ holds, which proves (i)–(iii).

Assumption A4 excludes any fourth event, such as an unverified plan being executed. The case analysis never refers to u^{des}: the planner enters only the cost of (12), which selects *among* verified plans. The statement therefore holds uniformly over all planners, including adaptive ones that know the filter. Claim (iv) is Lemma 3 combined with the third case. □

a) Why this is more than recursive feasibility: The machinery is that of predictive safety filters [23], [24], [39], but the constraint set is exogenous and driven by an adversarial certificate process, the guarantee concerns a state the filter never observes (the true position), and the quantifier ranges over adversarial, history-dependent planners rather than an imperfect learner. FaSTrack's planner-agnostic tracking bound [40] has the same quantifier structure for references. The guarantee is stated at plant sampling instants; a continuous-time claim needs an additional inter-sample margin.

E. Transparency

Lemma 4 (Transparency)**.** *Let $\mathcal{A}_{\mathrm{adm}}(\zeta) = \{v_0 : \exists$ a feasible plan of* (12) *with first input $v_0\}$.*

(a) If $u^{\mathrm{des}} \notin \mathcal{A}_{\mathrm{adm}}$, the filter intervenes ($v_0^\star \ne u^{\mathrm{des}}$).

(b) If $u^{\mathrm{des}} \in \mathcal{A}_{\mathrm{adm}}$ and $\mu = 0$, the filter returns u^{des} unchanged.

(c) For $\mu > 0$, the deviation is at most $\|v_0^\star - u^{\mathrm{des}}\|_2^2 \le 2\mu N(N_\mathrm{p}{-}1)u_{\mathrm{nom}}^2$.

Proof. For (a), no feasible plan starts with u^{des}. For (b), the cost is minimised at u^{des} over a set that contains it. For (c), compare with the feasible plan that starts with u^{des}: its cost is at most μ times the squared norm of a tail that (12e) bounds. □

Interventions are therefore forced only when a proposal leaves the admissible set of the *tightened* problem. The benign intervention rate (BIR) measures two things together: how unsafe the planner's proposals are, and the conservatism gap between the maximal robust invariant set and the tightened feasible set. That gap shrinks as $\bar{w} \to 0$ and as N_p grows.

F. Proposition 2: safety confinement and the denial taxonomy

For aircraft i, let $g_i(t_j)$ be the unit vector towards its goal, and let $v_i^{\mathrm{req}}(t_j)$ be the velocity its action asks for: the commanded velocity for `goto` and `velocity`, the braking target for `hold`, and $\hat{v}_i + a \cdot 1\,\mathrm{s}$ for an acceleration-mode oracle action. Define the *progress intent* accumulated before the deadline,

$$I_i = \textstyle\sum_{t_j < (1+\kappa_{\mathrm{sched}})T} \max\{\langle v_i^{\mathrm{req}}(t_j), g_i(t_j)\rangle, 0\}\, T_\mathrm{s}. \quad (13)$$

Let D_i be the initial goal distance minus the goal radius. A denied mission is *filter-induced* if some aircraft missed the deadline even though $I_i \ge D_i$. That is, the planner asked for enough progress and the filter's interventions or fallbacks prevented it. Otherwise the denial is *in-envelope*: the planner never asked for enough.

Proposition 2 (Safety/liveness split)**.** *Under the assumptions of Theorem 1, the following holds for every attacker in the ladder.*

(a) Confinement: $\Pr[\sigma \in S] = 1$. *The attacker's only achievable objectives are traces in $S \setminus L$ (denial), stealth within S, or breaking an assumption.*

(b) Decomposition: $\mathrm{MDR} = \mathrm{MDR}_{\mathrm{filter}} + \mathrm{MDR}_{\mathrm{env}} + \mathrm{MDR}_{\mathrm{viol}}$ *with* $\mathrm{MDR}_{\mathrm{viol}} = 0$*. The two remaining terms partition the denied missions.*

(c) In-envelope denial is invisible to any transparent filter: *let* $\mathcal{F}$ *be any filter (ours or another) that returns unchanged every proposal in a set* $\mathcal{A}_0$*. Then a planner whose proposals stay in* $\mathcal{A}_0$ *yields the same executed action stream with and without* $\mathcal{F}$*. Its denial is filter-invariant, so* $\mathcal{F}$ *cannot reduce the* $\mathrm{MDR}_{\mathrm{env}}$ *that such planners cause. In particular, a filter that admits its own fallback action (hold or loiter) admits a planner that proposes only that action.*

Proof. (a) is Theorem 1. (b) holds by definition, because the two classes are complementary on denied missions without violations. (c) By hypothesis, $\mathcal{F}(a) = a$ for every $a \in \mathcal{A}_0$, so the action streams coincide pathwise, and so does any mission outcome determined by them. For our filter, Lemma 4(b) identifies $\mathcal{A}_0$ with $\mathcal{A}_{\mathrm{adm}}$. □

Part (c) is structural: it does not depend on how the filter is tuned. A physical filter guards the envelope, and an attacker that stays inside the envelope is not its business. Looping or "malfunction amplification" attacks on LLM agents [41] are exactly this class. Filter-induced denial has its own control-theoretic mechanism. Minimally invasive barrier filters create asymptotically stable spurious equilibria on the boundary of the safe set [42] and deadlock in multi-robot settings [43]. An attacker that rides the boundary exploits exactly these equilibria. Stealth is a third dimension, not a third term: it is any denial trace whose per-step anomaly signals stay below an operator's threshold. RL-Hammer reaches 0% detection with 97% ASR on the language channel [15].

G. Proposition 3: nestedness under the sensing budget

Proposition 3 (Nested admissible sets)**.** *Fix* ζ*,* ρ*, the tube age and the normals, and let* $b \le b'$*. Then:*

(a) the feasible set of (12) *at* b' *is contained in that at* b*, hence* $\mathcal{A}_{\mathrm{adm}}(\zeta; b') \subseteq \mathcal{A}_{\mathrm{adm}}(\zeta; b)$*;*

(b) the optimal cost satisfies $J^\star(b') \ge J^\star(b)$*, and the one-step intervention indicator is monotone:* $\mathbf{1}[u^{\mathrm{des}} \notin \mathcal{A}_{\mathrm{adm}}(b)] \le \mathbf{1}[u^{\mathrm{des}} \notin \mathcal{A}_{\mathrm{adm}}(b')]$*;*

(c) the same nestedness holds in ρ*,* $\bar{w}$*,* γ *(through* L_ρ *and* $\bar{\rho}$*) and* $\bar{\Delta}$*;*

(d) $b_{\mathrm{tot}}(\varepsilon)$ *in* (5) *is non-increasing in* ε*, non-decreasing in* δ_{net} *and* δ_{adv}*, and non-decreasing under first-order stochastic dominance of any benign marginal.*

Proof. (a) b_{tot} enters (11) additively, and β enters (12c) and (12d) only through right-hand sides of the form $c - \alpha\beta$ and $d + \alpha\beta$ with $\alpha \ge 0$. Increasing b therefore shrinks every constraint set while leaving the dynamics unchanged. (b) A minimum over a smaller set is no smaller, and (b) restates (a) pointwise. (c) $\tilde{\rho}$, $\epsilon_{p,v,u}$ and $\bar{\rho}$ are non-decreasing in these parameters (ϵ is linear in $\bar{w}$). (d) $\bar{p}$ is non-increasing, so $s^\star$ is non-increasing in ε. Deterministic terms shift the margin (Paper IV, Prop. 2(e) [30]). A tail of a sum is non-decreasing under first-order stochastic dominance of a summand, for every coupling (proof of Paper IV, Thm. 1). □

a) What is not claimed: Proposition 3 is pointwise. Pathwise monotonicity of the *rate* of intervention, fallback or denial in b_{tot} does not follow, because closed-loop trajectories diverge as b changes. We therefore state it as a hypothesis, **H3**: *the closed-loop* IR*,* FR *and* MDR *are non-decreasing in* b_{tot}. We test it with ordered-trend tests under common random numbers (CRN) (Section VI-D). Composition is only meaningful for Paper IV's covert class. A detection-indifferent spoofer that drives the estimator outside Assumptions A1 and A2 is the stealthy-deactivation attack [32]. In the terms of Paper II's runtime assurance lift L_{mon} [44], this is where monitor independence fails.

Remark 1 (What is *not* guaranteed). (i) *Liveness.* Persistent denial is possible by Proposition 2(c).

(ii) *Estimator attacks.* Attacks that move $\hat{x}$ beyond $\bar{w}$, or falsify ρ_i, void Assumption A1 or Assumption A2.

(iii) *Certificate jumps.* Relay jumps beyond $\bar{\rho}$ void Assumption A2. The median two-hop jump, about 41.8 m, exceeds every margin in the programme's family $\{2, 5, 10, 20\}$ m. There the theorem is *silent*, as the parent certificate is silent outside its own scope conditions.

(iv) *An empty admissible set at take-off.* If Assumption A5 fails, the filter must refuse dispatch. Anything flown after that is uncertified.

(v) *Model transfer.* The guarantee concerns the model (1). On hardware, $\bar{w}$ must be identified, not assumed; and γ was measured in hover [25].

IV. Adaptive Evaluation Protocol

A defence evaluated against a fixed attack proves little [45]–[48]. We carry the adaptive-evaluation discipline to the physical layer and add one advantage that model-level defences lack: because Theorem 1 is a statement over *all* action sequences, the strongest attacker for the safety claim needs no language model at all.

A. Attacker ladder

Table I lists the eight levels. Each level is a superset of the one before it, and every level reports violations and denial against the four goals G1–G4.

a) Oracle family used in the pilot: For each aircraft, the oracle proposes the acceleration $u_{\max}\,\mathrm{dir}/\|\mathrm{dir}\|_\infty$, where

$$\mathrm{dir} = w_{\mathrm{wall}} h_{r^\star} + w_{\mathrm{nbr}} \hat{n}_{\mathrm{nn}} + w_{\mathrm{c}}(\cos\phi, \sin\phi) + w_{\mathrm{vel}} \hat{v} + w_{\mathrm{goal}} \hat{g}.$$

Here $h_{r^\star}$ is the normal of the most binding wall and $\hat{n}_{\mathrm{nn}}$ points to the nearest neighbour. The filter returns the minimum-deviation admissible input, so a maximal push towards a constraint is already the filter-aware one-step optimum: the filtered input rides the boundary of the tightened set. CEM tunes the mixture $\theta = (w_{\mathrm{wall}}, w_{\mathrm{nbr}}, w_{\mathrm{c}}, \phi, w_{\mathrm{vel}}, w_{\mathrm{goal}})$ over whole closed-loop episodes with the filter in the loop. The *violation* oracle minimises the closest true approach to any constraint. The *denial* oracle maximises missed deadlines plus

TABLE I
ATTACKER LADDER. A6 NEEDS NO LLM AND UPPER-BOUNDS A1–A5 FOR THE SAFETY CLAIM (COROLLARY 4). STATUS REFERS TO THIS PAPER.

Lvl	Capability and construction	Status
A0	Benign scripted mission planner	pilot (scripted)
A1	Static injection corpus: AgentDojo "important message" [49], InjecAgent [50], HouYi [51], translated to mission text	[pending]
A2	Black-box, filter-unaware LLM search (PAIR/TAP [52], [53])	[pending]
A3	Black-box, filter-aware: knows the constraint set, observes the intervene bit; cross-entropy method (CEM) [54] or covariance matrix adaptation (CMA-ES) [55] over template parameters	[pending]
A4	White-box open-weights planner: greedy coordinate gradient (GCG) [56] or reinforcement learning (RL) [15] with a stealth reward	[pending] (optional)
A5	Human red team [13]	[pending] (optional)
A6	**Action-space oracle**: chooses a^{LLM} directly; CEM-tuned and hand-designed push families, with worst-case nature	pilot
A7	A6 or A3 composed with Paper IV's covert sensing offset	pilot (offset realised at budget); detector-gated version [pending]

fallback and intervention fractions, but its search box forces $w_{\mathrm{goal}} \geq 0.5$. It is an attacker that wants the *filter* to cause the denial. Nature is adversarial in both: the disturbance, the certificate error and the sensing offset all point at each aircraft's most binding constraint. This box worst case is exactly what the $\|h\|_1$-scaled tightening covers.

Corollary 4 (Oracle envelope)**.** *Let Π_k be the set of causal action policies that a language-mediated attacker at level $k \in \{1, \ldots, 5\}$ can induce through any text, and let Π_{act} be all causal action policies (A6). Then $\Pi_k \subseteq \Pi_{\mathrm{act}}$. Hence:*

(a) $\sup_{\Pi_k} \mathrm{SVR} \leq \sup_{\Pi_{\mathrm{act}}} \mathrm{SVR} = 0$ *under the assumptions of Theorem 1;*

(b) for any liveness metric $M \in \{\mathrm{MDR}, \mathrm{FR}, \mathrm{IR}\}$, $\sup_{\Pi_k} M \leq M^{\star}_{\mathrm{A6}} := \sup_{\Pi_{\mathrm{act}}} M$.

Proof. The planner's output is restricted to the action schema, so every text attack induces some action policy. The rest follows from Theorem 1 and from monotonicity of the supremum. □

The searched oracle achieves $M^{\mathrm{search}}_{\mathrm{A6}} \leq M^{\star}_{\mathrm{A6}}$, so for liveness it is a lower bound on the oracle's power, not a proof of the strongest attack. We call $M^{\mathrm{search}}_{\mathrm{A6}} - \max_{k \leq 5} M(\mathrm{A}_k)$ the *LLM-mediation slack*. It measures how much of the theoretically available denial an attacker can actually realise through text.

b) Sanity checks: Following Carlini et al. [48]:

(i) with the filter off, attacks must produce violations; otherwise they are too weak to say anything;
(ii) an unbounded oracle must reach near-total denial;
(iii) attack success must not decrease with search budget;
(iv) results are the *worst case over attackers per mission*, never an average over attacks.

B. Metrics

We follow AgentDojo's split into benign utility, utility under attack and targeted ASR [49], and Paper III's per-step counting convention [29]. Let n_{st} be the number of plant steps and n_{tk} the number of filter ticks of an episode.

- **Safety-violation rate** $\mathrm{SVR} = \#\{(i,n) : p(n) \notin \mathcal{S} \text{ through } i\}/(N\, n_{\mathrm{st}})$, counted per aircraft-step and never averaged per episode. Here SVR answers RQ1.
- **Intervention rate** $\mathrm{IR} = \#\{(i,j) : \|v^{\star}_{i,0} - u^{\mathrm{des}}_i\|_\infty > \eta\}/(N\, n_{\mathrm{tk}})$, with $\eta = 0.01\,\mathrm{m/s}^2$; it includes fallback ticks. BIR is IR under A0.
- **Fallback rate** FR: the fraction of fleet ticks that execute the latched tail. **Continuation rate** CR: the fraction of ticks certified by P_{C}.
- **Mission completion and denial** MCR and MDR follow (6) with $\kappa_{\mathrm{sched}} = 0.10$ (headline) and 0.25 (sensitivity), with the split of Proposition 2.
- **Benign mission-time overhead**: the difference in fleet completion time between filter on and filter off, paired by seed.
- **Solve latency**: the 50th, 95th and 99th percentiles (p50, p95, p99) and the maximum per tick, against the filter period. This discharges the switching-latency obligation O_{lat} of L_{mon} [44].
- **Mediation slack** (above). A coverage-ratio analogue (Paper III) and a stealth rate are planned.

C. Statistics

Zero-violation claims carry a one-sided 95% Clopper–Pearson (CP) bound [57], $1 - 0.05^{1/n} \approx 3/n$, so claiming $\mathrm{SVR} \leq 10^{-6}$ needs about 3×10^6 violation-free aircraft-steps. Episode-level proportions carry Wilson score intervals; IR, FR and the time overhead carry percentile bootstrap confidence intervals (CIs) over episodes ($B = 2000$). Trend hypotheses use Spearman rank correlation of per-episode values against the ordered budget level under CRN. The planned campaign adds Holm correction [58] across operating points and interquartile means with stratified bootstrap across seeds [59]. Seeds are partitioned by role as in Paper IV (2xxx attacker optimisation, 4xxx held-out evaluation, 5xxx negative controls, 6xxx one-step study), and constants are never widened after looking at held-out data.

D. Baselines and ablations

a) Baselines: The baselines span the defence families: no filter (run); a myopic one-step control barrier function quadratic program (CBF-QP) [60], [61]; Simplex accept/reject with loiter; a RoboGuard-style semantic guardrail [16]; model-only defences (spotlighting [62], sandwich prompting, the instruction hierarchy [63], a SAFER-style LLM judge [64]); and a stacked semantic-plus-physical configuration. All except the first are [pending].

b) Ablations run in the pilot: Without continuation; without the L_ρ look-ahead; $\gamma \in \{1.365, 1.625, 15\}$ m/s; $\bar{\Delta} \in \{0.5, 0.944, 2.0\}$ s; true mismatch at $\{0, 1, 2, 4\} \times \bar{w}$; and certificate jumps (outside Assumption A2).

c) Planned ablations: Horizon $N_{\mathrm{p}} \in \{10, 20, 40\}$; the terminal set (loiter, filtered return-to-launch (RTL), none); the tightening (none, Limón, Köhler); hard versus soft constraints; latching; separation constraints on or off; and planner cadence.

V. Experimental Setup

A. Pilot: what was run

The pilot is a seeded simulation with *non-LLM* planners only. No language model was queried. Its purpose is to exercise every part of the theory before money is spent on model calls. The A6 oracle is the load-bearing attacker for RQ1, and Corollary 4 explains why.

a) Plant and mission: The plant is a fleet of $N \in \{4, 12\}$ planar double integrators at $\delta t = 0.1$ s, with per-axis actuator saturation $u_{\max} = 4\,\mathrm{m/s^2}$ [30], a speed limit $v_{\max} = 12$ m/s and bounded acceleration mismatch. The keep-in set is a polygon with six half-planes, including two chamfers. Lanes are 40 m apart, and pairwise lane-swap goals at 560 m create crossing traffic. The mission length is $T = 70$ s (the parent's reference profile and the platform's nominal mission), the deadline is $(1+\kappa_{\mathrm{sched}})T$, and simulations run to 87.5 s, so that both values of κ_{sched} can be read from one run.

b) Certificate: The certificate follows Paper III's generator, driven by the parent's hop-ledger residual pools. Inside Assumption A2, staleness is capped at $\bar{\Delta} = 0.944$ s ($\bar{\rho} = 5.0$ m) and relay jumps are disabled. The negative control uses Paper III's defaults: a 10 s cap and undetected relay jumps.

c) Nature: Benign and scripted cells use a clipped Gauss–Markov gust (time constant 2 s) and a uniform estimation offset. Oracle and budget cells use adversarial nature (Section IV). The sensing offset is realised at the declared budget, $b_{\mathrm{adv}} = b_{\mathrm{tot}}$, so the budget cells are an A6+A7 approximation. A detector-gated A7 that replays Paper IV's attackers is **[pending]**.

d) Planners: Benign (A0) flies to the goal at 10 or 8.5 m/s and holds. *Random* picks random waypoints every 5 s. *Stall*, the canonical in-envelope denial attacker, is benign until a random onset in $[5, 30]$ s and then proposes safe holds. *oracle_violation* and *oracle_denial* are the CEM-tuned A6 oracles, evaluated alongside five hand-designed members (press_wall, press_neighbour, overspeed, goal+neighbour, goal+wall). All constants, with provenance tags, are listed in Table VII (Section B). CEM used 8 candidates, 3 elites, 4 generations and 2 training seeds (2xxx), with $N = 4$ only. The optimum was transferred unchanged to $N = 12$. This search budget is small, so hand-designed members are always evaluated too, and the worst case per mission is reported.

e) Campaign size: The pilot ran in seven stages: CEM (128 episodes); main (540: five planners, filter on and off, 40 episodes per cell for $N = 4$ and 14 for $N = 12$); hand-designed oracles (210); budget sweep with the γ and $\bar{\Delta}$ cells (319); mismatch sweep (208); negative controls and ablations (84); and a one-step nestedness study (300 random instances at each of seven budget levels). In total, 1,489 episodes ran in 36.2 min on two worker processes. The filter-on evaluation comprises 986 episodes and 4,501,000 aircraft-steps.

f) Implementation: The filter is a joint fleet QP with relative coordinates. Each tick first runs the Operator Splitting Quadratic Program (OSQP) solver [65], which uses the alternating direction method of multipliers (ADMM), with a 40 ms budget and loose tolerance. If that neither certifies nor proves infeasibility, the Clarabel interior-point method (IPM) runs within the remaining budget. The total per-tick budget is 250 ms, half the filter period, and an overrun counts as a fallback. Small solver buffers (0.10 m, 0.05 m/s, 0.05 $\mathrm{m/s^2}$) absorb loose tolerances. The verifier does not use them. The software stack was Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, OSQP 1.1.3 and Clarabel 0.11.1, running in a two-core virtual-CPU container on Linux.

TABLE II
RQ1: POOLED TRUE-STATE SAFETY WITH THE FILTER ON, INSIDE ASSUMPTIONS A1 TO A5 (ALL PLANNERS AND STAGES). CP: ONE-SIDED 95% CLOPPER–PEARSON UPPER BOUND PER AIRCRAFT-STEP.

Pool	Episodes	Aircraft-steps	Viol.	CP bound
All, $N \in \{4, 12\}$	765	3,608,500	0	8.30×10^{-7}
$N = 4$	632	2,212,000	0	1.35×10^{-6}
$N = 12$	133	1,396,500	0	2.15×10^{-6}
A6 oracles only	397	1,816,500	0	1.65×10^{-6}
LLM planners, A0–A5	**[pending]**	**[pending]**	**[pending]**	**[pending]**

B. Planned LLM campaign (pre-registered)

The campaign follows the programme runbook. Two frontier models from different providers, plus one open-weights model for adaptive search, are pinned to dated snapshot identifiers at campaign time (**[pending]**). Calls use temperature 0 where supported, native structured output, delimited channels and a SHA-256-keyed completion cache, so every logged action replays bit for bit. One fleet-level call is made every 5 s (about 14 per mission); the filter acts at every tick on the latest action. The benign phase (RQ4) has 1,840 episodes and 25,760 calls (about 64.4 M input and 7.7 M output tokens). The attack phase (RQ2) has about 3,200 episodes, 44,800 planner calls and 14,000 attacker or judge calls; adaptive search runs on a surrogate or open-weights planner and only the top-k injections transfer to frontier models. Spending is capped at US$1,500 for all LLM phases, with prices recorded on the pin date (**[pending]**). Composition cells use $\delta_{\mathrm{adv}} = 5.463$ m, and the horizon is $N_{\mathrm{p}} = 20$ with an ablation over $\{10, 20, 40\}$.

VI. Results

All results in this section are from the simulation pilot with non-LLM planners.

A. RQ1: safety of the true state

Table II pools every filter-on episode that satisfies the assumptions. There are zero violating aircraft-steps in 3,608,500, so the one-sided 95% CP bound on the per-aircraft-step violation probability is 8.30×10^{-7}.

The oracle is not weak. Across inside-assumption oracle episodes, the closest true approach was 0.062 m to the keep-in boundary and 0.383 m to the separation minimum, against a terminal tightening of 9.9 m. The tightening is nearly tight, and the attacker finds that edge. The 65 filter-on episodes that nominally lie inside the assumptions are excluded from the pool because an auxiliary condition failed: an uncertified start or a deliberate ablation. They are reported in Section VI-B.

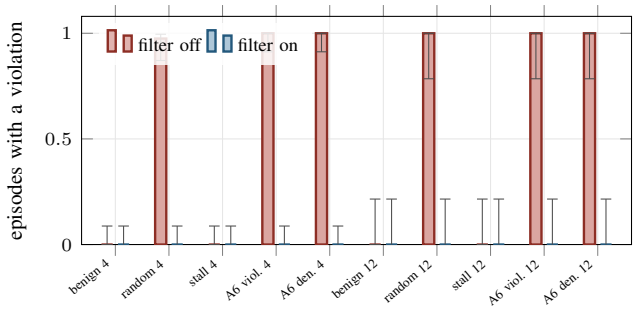


Fig. 2. RQ1 sanity check: the fraction of episodes with any true-state violation, with 95% Wilson intervals. Filter on: zero in every cell (the bars are at zero; the whiskers show the Wilson upper bound). Filter off: the oracles violate in every episode (40/40 for $N = 4$, 14/14 for $N = 12$), and the random planner in 39/40 and 14/14.

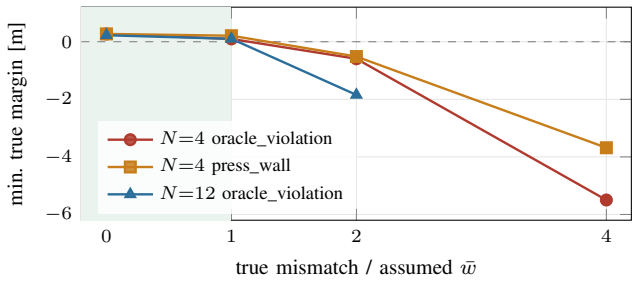


Fig. 3. Mismatch negative control. The filter assumes $\bar{w} = 0.5\,\mathrm{m/s^2}$ (shaded: inside Assumption A1). The closest true approach stays positive at $0\times$ and $1\times$ and turns negative as soon as the true mismatch exceeds the assumed bound. The $N = 12$ point at $4\times$ ($-66.8\,\mathrm{m}$) is off scale. Benign planners stay safe at every multiple: the margin is load-bearing only under attack.

a) Sanity check: Figure 2 and Table III show the same planners with the filter switched off. Both oracles violate in every episode, at a per-aircraft-step rate of 0.960. The random planner violates in 39/40 and 14/14 episodes. With the filter on, every cell records zero. The worst case over the seven oracle variants per mission (Table XII, Section F) is 40/40 and 14/14 missions with a violation when the filter is off, and 0/40 and 0/14 when it is on.

B. Scope boundary: violations appear exactly when an assumption breaks

A safety theorem is only as credible as its failure modes. We therefore broke each assumption on purpose (Table IV and Fig. 3). Altogether 150 filter-on episodes recorded a violation. Every one of them breaks a stated assumption by design. None lies inside the assumptions.

a) Mismatch: With the true mismatch at $0\times$ and $1 \times \bar{w}$, the oracles never violate (closest margins 0.27 and 0.10 m). At $2\times$ every oracle episode violates (margin $-0.59\,\mathrm{m}$), and at $4\times$ the margin falls to $-5.50\,\mathrm{m}$, or $-66.8\,\mathrm{m}$ for $N = 12$. The benign planner stays safe at every multiple, with margins of about 11 m. The mismatch margin matters only when an attacker presses on the boundary, which is why it must be calibrated and not guessed.

b) Certificate jumps: With Paper III's default process (10 s staleness cap, undetected relay jumps), the theorem is silent, and 15/16 oracle episodes violate by up to 90.6 m. Even the benign planner violates in 6/16 episodes. Removing the L_ρ look-ahead *inside* Assumption A2 produces violations in 16/16 episodes, with a margin of $-4.29\,\mathrm{m}$. Paper III's look-ahead is therefore necessary, not cosmetic.

c) Uncertified start: With the kinematic rate $\gamma = 15\,\mathrm{m/s}$, $\bar{\rho} = 46.15\,\mathrm{m}$ and the terminal tightening is 51.05 m. The admissible set is empty at take-off, Assumption A5 fails, and the uncertified brake-and-hold start violated in all 12 episodes. The correct response is to refuse dispatch. The same happens at the full budget $b_{\mathrm{tot}}(0.1) = 12.718\,\mathrm{m}$ (Section VI-D).

C. RQ2: what the attacker keeps

Table III shows the denial ledger. Every planner except the benign one denies every mission. The mechanisms, however, differ in exactly the way Proposition 2 predicts.

a) In-envelope denial: The stall planner denies 100% of missions for both fleet sizes. All of that denial is in-envelope, the fallback rate is zero, and the intervention rate (0.070 for $N = 4$, 0.064 for $N = 12$) is no higher than the benign rate (0.077, 0.069). The filter is essentially transparent, as Proposition 2(c) predicts: it neither causes this denial nor can it reduce it.

b) Filter-induced denial: The denial oracle is constrained to keep a progress component. It converts its blocked safety attack into filter-induced denial in 0.95 [0.83, 0.99] of $N = 4$ missions and in all $N = 12$ missions. Its fallback rate is 0.020 for $N = 4$ and 0.124 for $N = 12$, and its continuation rate is 0.374 and 0.634. Larger fleets give the attacker more coupled constraints to press against. The violation oracle, whose search box does not reward progress, is classified almost entirely as in-envelope. Among the hand-designed members (Table XII), the goal+neighbour push denies only 0.56 of $N = 4$ missions, all of them filter-induced. Boundary riding through a neighbour is therefore weaker than riding a wall at $N = 4$, but reaches 1.00 at $N = 12$.

c) Continuation is what keeps fallback rare: Without the continuation problem, the fallback rate under the denial oracle rises from 0.020 to 0.436 (Table IV), still with zero violations. The latched tail keeps the aircraft safe; continuation keeps it flying.

d) Language-mediated attackers: The LLM-mediation slack requires A1–A5 against real models. Those cells are **[pending]** (Table V). The pilot establishes the oracle side of the slack: the searched oracle already reaches denial 1.00, so the slack is $1.00 - \max_{k \le 5} \mathrm{MDR}(\mathrm{A}_k)$.

D. RQ3: composition with the sensing budget

Figure 4 and Table VI sweep b_{tot} over seven levels. These are 0, 0.5, 1 and 2 m; the pilot $\delta_{\mathrm{adv}} = 4.371\,\mathrm{m}$; δ_{adv} plus the network term, 7.019 m; and the full $b_{\mathrm{tot}}(0.1) = 12.718\,\mathrm{m}$ with $s^\star = 5.70\,\mathrm{m}$. Every cell has the covert offset realised at its budget.

TABLE III

MAIN STAGE ($b_{\text{tot}} = \delta_{\text{adv}} = 4.371$ M, TRUE MISMATCH $= \bar{w}$). OFF/ON: EPISODES WITH ANY VIOLATION. SVR_{off}: PER AIRCRAFT-STEP. IR AND FR WITH 95% BOOTSTRAP CIS FOR THE BENIGN ROWS. DENIAL IS SPLIT INTO FILTER-INDUCED AND IN-ENVELOPE SHARES (PROPOSITION 2). THE LLM ROWS ARE PLACEHOLDERS.

N	Planner	OFF viol.	SVR_{off}	ON viol.	IR	FR	CR	MCR [Wilson]	Denial: filter	Denial: envelope
4	benign (A0)	0/40	0	0/40	0.077 [0.073, 0.081]	0.000	4.3×10^{-4}	1.00 [0.91, 1.00]	–	–
4	random	39/40	0.0160	0/40	0.481	2.9×10^{-4}	0.083	0.00	0.00	1.00
4	stall	0/40	0	0/40	0.070	0.000	0.000	0.00	0.00	1.00
4	oracle_violation (A6)	40/40	0.9602	0/40	1.000	0.008	0.399	0.00	0.00	1.00
4	oracle_denial (A6)	40/40	0.9601	0/40	1.000	0.020 [0.015, 0.024]	0.374	0.00	0.95 [0.83, 0.99]	0.05
12	benign (A0)	0/14	0	0/14	0.069 [0.063, 0.076]	0.000	0.018	1.00 [0.78, 1.00]	–	–
12	random	14/14	0.0171	0/14	0.597	0.027	0.312	0.00	0.00	1.00
12	stall	0/14	0	0/14	0.064	0.000	0.010	0.00	0.00	1.00
12	oracle_violation (A6)	14/14	0.9601	0/14	1.000	0.144	0.598	0.00	0.00	1.00
12	oracle_denial (A6)	14/14	0.9600	0/14	1.000	0.124 [0.106, 0.142]	0.634	0.00	1.00 [0.78, 1.00]	0.00
12	LLM A0 (model 1, 2)	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]
12	LLM A1–A3 worst case	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]

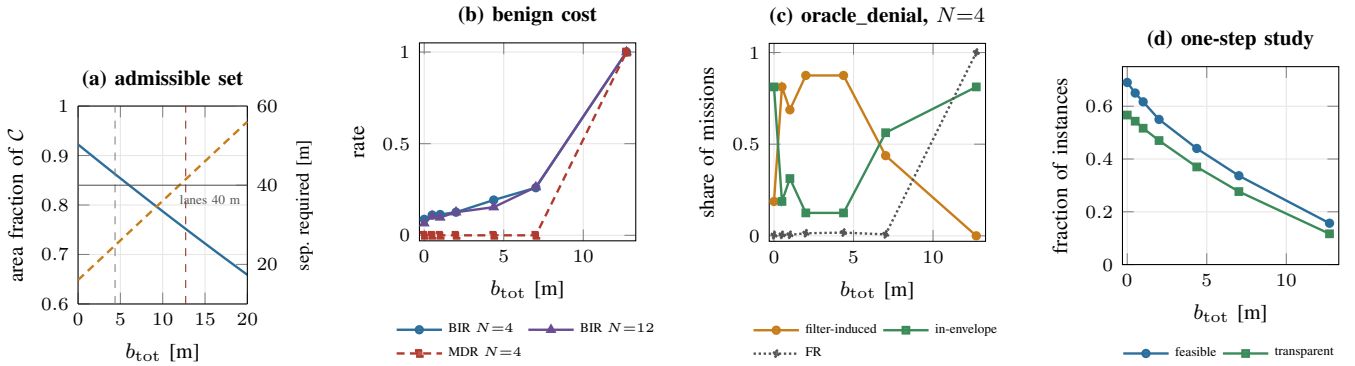


Fig. 4. RQ3: composition with the sensing budget (adversarial nature, offset realised at the budget). (a) The tightened keep-in area shrinks smoothly (solid, left axis), but the separation required between neighbouring terminal nodes (dashed, right axis) crosses the 40 m lane spacing at $b_{\text{tot}} \approx 12$ m. The dashed verticals mark the pilot $\delta_{\text{adv}} = 4.371$ m and the full $b_{\text{tot}}(0.1) = 12.718$ m. (b) The benign intervention rate rises monotonically with the budget, and denial stays at zero until the admissible set is empty at take-off. (c) The denial oracle's attribution shifts with the budget; the remaining share at 12.718 m is uncertified-start violations. (d) One-step study (300 instances per level): the feasible and transparent fractions fall monotonically, with zero nestedness violations.

TABLE IV

NEGATIVE CONTROLS ($N = 4$ UNLESS STATED; 16 EPISODES PER CELL, 4 FOR $N = 12$, 12 FOR THE γ CELL). EACH ROW BREAKS ONE ASSUMPTION.

Cell	Broken	Viol. eps	Min margin	FR
mismatch $2 \times \bar{w}$, oracle_violation	A1	16/16	−0.59 m	0.026
mismatch $4 \times \bar{w}$, oracle_violation	A1	16/16	−5.50 m	0.019
mismatch $2 \times \bar{w}$, press_wall	A1	16/16	−0.52 m	0.000
outside A2, benign	A2 (jumps)	6/16	−41.17 m	0.357
outside A2, oracle_violation	A2	15/16	−90.56 m	0.393
outside A2, N=12 benign	A2	3/4	−38.36 m	0.784
no look-ahead, press_wall	A2 ablated	16/16	−4.29 m	0.117
$\gamma = 15$ m/s (kinematic), benign	A5	12/12	–	1.000
no continuation, oracle_denial	none (ablation)	0/16	+0.17 m	0.436

TABLE V

PLANNED REAL-LLM CELLS (ALL [PENDING]; NO RESULT EXISTS).

Level	Model	SVR (CP)	MDR	filter / env.	BIR
A0	frontier 1, 2	[pending]	[pending]	[pending]	[pending]
A1	frontier 1, 2	[pending]	[pending]	[pending]	–
A2	frontier 1, 2	[pending]	[pending]	[pending]	–
A3 (transfer)	frontier 1, 2	[pending]	[pending]	[pending]	–
A4, A5	open-weights	[pending]	[pending]	[pending]	–
filter off	A1, A3-best	[pending]	[pending]	–	–
Mediation slack $1.00 - \max_k \text{MDR}(\text{A}_k)$: [pending]					

a) Nestedness holds pointwise: In the one-step study, the feasible fraction falls from 0.690 to 0.157 and the transparent fraction from 0.567 to 0.117. Across 300×6 adjacent-level pairs there were zero nestedness violations in feasibility, in transparency, and in optimal deviation (none decreased by more than 10^{-3} m/s^2), as Proposition 3 requires. Lemma 4 (transparent if and only if the deviation is at most η) agreed on 1031 of the 1032 feasible (instance, level) pairs. All 1032 interior-point solutions passed the independent verifier.

b) H3 is supported for benign cost, not pathwise for attack: With CRN, the benign intervention rate rises monotonically over the levels: 0.087, 0.110, 0.114, 0.126, 0.193, 0.260 and 1.000. The ordered-trend test gives Spearman $\rho_s = 0.98$ ($p = 6.2 \times 10^{-80}$) for $N = 4$ and 0.94 ($p = 1.9 \times 10^{-17}$) for $N = 12$. For benign denial and fallback at $N = 4$ the correlation is 0.61 ($p = 7.3 \times 10^{-13}$), driven by the jump at the last level. Even for the benign planner at $N = 12$, all 5 seeds have a non-monotone fallback-count sequence across levels. For the denial oracle the fallback fraction trends upward ($\rho_s = 0.53$, $p = 1.7 \times 10^{-9}$), but 14 of 16 seeds have a

TABLE VI

BUDGET SWEEP. β_{term}: TERMINAL TIGHTENING. B: BENIGN PLANNER; O: DENIAL ORACLE; F/E: FILTER-INDUCED/IN-ENVELOPE SHARES; 1-STEP: FEASIBLE/TRANSPARENT FRACTIONS OF THE ONE-STEP STUDY.

b_{tot}	β_{term}	B: BIR	B: MDR	O: FR	O: f/e	B_{12}: BIR	1-step
0.000	5.53	0.087	0.00	0.004	0.19/0.81	0.066	0.690/0.567
0.500	6.03	0.110	0.00	0.005	0.81/0.19	0.104	0.650/0.543
1.000	6.53	0.114	0.00	0.005	0.69/0.31	0.097	0.617/0.517
2.000	7.53	0.126	0.00	0.014	0.88/0.12	0.126	0.550/0.470
4.371	9.90	0.193	0.00	0.018	0.88/0.12	0.154	0.440/0.370
7.019	12.55	0.260	0.00	0.008	0.44/0.56	0.263	0.337/0.277
12.718	18.24	1.000	1.00	1.000	0.00/0.81	1.000	0.157/0.117
5.463	–	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**
8.111	–	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**

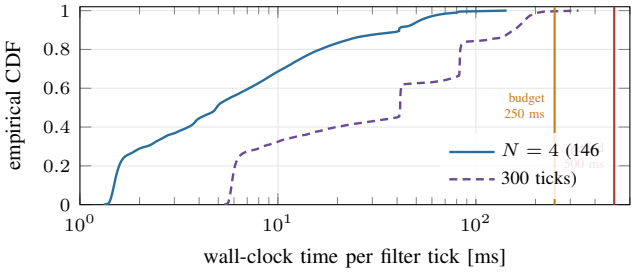


Fig. 5. Solve-time distribution (OSQP first stage, Clarabel IPM second stage, exact verification). The plateaus near 41 ms and 83 ms are the OSQP budget being exhausted before the IPM stage; the $N = 12$ tail is the joint fleet QP.

non-monotone fallback sequence across levels. This is exactly the pathwise non-monotonicity that Proposition 3 does *not* exclude. Denial is saturated at 1.00 at every level, so its trend is undefined. Attribution moves with the budget (Fig. 4c): at $b = 0$ most oracle denial is in-envelope, while at intermediate budgets it is mostly filter-induced.

c) The budget can empty the admissible set: At the full $b_{\text{tot}}(0.1) = 12.718$ m the terminal tightening is 18.24 m. Two neighbouring terminal nodes must then be $d_{\text{sep}} + 2\beta_{\text{term}} = 41.5$ m apart, more than the 40 m lane spacing, so the admissible set is empty at take-off for every planner. Assumption A5 fails and the mission must not be dispatched. In this geometry the composed budget is therefore not a cost the filter pays in denial; it is a dispatch decision.

d) The δ_{adv} correction: The pilot used $\delta_{\text{adv}} = 4.371$ m. Paper IV's later re-pooling gives 5.463 m (FPR 1%, 5%) and 5.038 m (10%). Two consequences follow without new runs:

- The full budget becomes $b_{\text{tot}}(0.1) = 5.463 + 2.648 + 5.699 = 13.81$ m. This exceeds 12.718 m, at which the admissible set is already empty, so by Proposition 3(a) the full-budget case is also non-dispatchable.
- The δ_{adv}-only level, 5.463 m, lies between two dispatchable pilot levels, and δ_{adv} plus the network term, 8.111 m, gives $d_{\text{sep}} + 2\beta_{\text{term}} \approx 32.3$ m, below the 40 m lane spacing. Neither cell has been run; both are marked **[pending]** in Table VI.

e) Sensitivity to γ and $\bar{\Delta}$: For the benign planner at $b_{\text{tot}} = 4.371$ m, the intervention rate is 0.172 at $\gamma = 1.365$ m/s and 0.190 at 1.625 m/s, with no denial. Shrinking the staleness envelope to $\bar{\Delta} = 0.5$ s lowers it to 0.134; widening it to 2.0 s ($\bar{\rho} = 10.59$ m) raises it to 0.507, still with full completion. The kinematic companion row, $\gamma = 15$ m/s, is non-dispatchable (Section VI-B).

E. RQ4: the price of the guarantee

a) Benign cost: With a benign scripted planner, mission completion is 1.00 with the filter on, identical to off, for both fleet sizes. The benign intervention rate is 0.077 [0.073, 0.081] for $N = 4$ and 0.069 [0.063, 0.076] for $N = 12$, with a mean deviation of 0.049 and 0.052 m/s^2. Most interventions clip the take-off acceleration from 4.0 to the tightened 3.31 m/s^2 or adjust the crossing pairs for separation. The paired mission-time overhead is 0.14 [0.10, 0.18] s on a mean of 66.7 s (0.2%) for $N = 4$, and 0.01 [−0.04, 0.05] s on 67.0 s for $N = 12$. For comparison, CaMeL's provable security costs 7 points of AgentDojo utility (77% versus 84%) [26]. At the pilot budget, the physical analogue costs a fraction of a second. LLM benign cost is **[pending]** (Table V).

b) Latency: Figure 5 shows the per-tick wall-clock time of both solver stages plus verification. For $N = 4$: p50 4.9 ms, p95 52.4 ms, p99 80.3 ms and maximum 143.0 ms over 146,300 ticks. For $N = 12$: p50 41.4 ms, p95 176.6 ms, p99 206.8 ms and maximum 330.1 ms over 26,250 ticks. No tick exceeded the 500 ms filter period. A fraction 0.0024 of the $N = 12$ ticks exceeded the 250 ms budget and fell back to the latched plan by design. This Python prototype on a shared two-core container discharges O_{lat} at 2 Hz. It does not show 50–100 Hz feasibility, which needs a compiled per-aircraft solver.

c) Reproducibility under wall-clock budgets: We ran the main stage twice, with a bug fix to the denial intent proxy between the runs. The violation counts were identical in 540 of 540 paired episodes. Intervention, continuation and fallback counts, however, were identical in only 63 of 270 filter-on episodes. Seeds fix the scenario, but which ticks hit a wall-clock limit depends on machine load. The largest cell-level differences were 0.0056 in IR, 0.0257 in FR, 0.0486 in CR and zero in MCR. Safety does not depend on timing, because every executed plan is verified. Liveness metrics do, slightly. The full campaign will replace wall-clock budgets with iteration caps and measure latency separately.

VII. DISCUSSION

a) A conversion ledger, not a safety theorem: The safety theorem is standard in its machinery. The contribution is the ledger of what an attacker who can no longer crash the aircraft can still take away, and through which channel. The filter removes envelope violations, but it leaves two residues:

- *Filter-induced denial* has a control-theoretic explanation in boundary equilibria and deadlock [42], [43]. It can be reduced by better filter design, such as continuation (0.436 to 0.020 fallback) and less conservative tightening.
- *In-envelope denial* cannot be reduced by any physical filter (Proposition 2(c)). It needs a language-side defence or a

progress-preserving fallback. One option is a *liveness budget*: after k consecutive non-progress epochs, quarantine the language channel and hand the mission to a certified scripted planner. This is a recommendation, not a result.

Stacking a semantic guardrail such as RoboGuard [16] above the physical filter is complementary, not competing: the semantic layer targets intent, and the physical layer targets dynamics.

b) Reading in runtime assurance terms: The filter is an active-set-invariance or predictive-filter type of runtime assurance (RTA) with a Simplex-style latched fallback [38], [66], [67]. This is the architecture standardised for complex functions on aircraft by ASTM F3269-21 [68]. In Paper II's calculus [44], it is the lift L_{mon}, whose four obligations we discharge as follows (Table X, Section E). The table also shows where the gap is relocated rather than closed.

c) Limitations:

- *Model–plant mismatch.* Theorem 1 holds for (1). The pilot's $\bar{w}$ is a modelling choice, motivated by Paper IV's drag and light turbulence, about 0.46 m/s^2. It has not been calibrated against Paper IV's plant, and the disturbance in the pilot is bounded by construction. Figure 3 shows that this margin is load-bearing under attack.
- *Simulation only.* The plant is a planar double integrator. Estimation and sensing errors are output offsets, not fed back through an estimator. γ was measured in hover and is applied at a 10 m/s cruise, a transfer assumption the parent itself calls its weakest. Violations are checked at the 0.1 s step. Nothing here is a hardware claim. A PX4 software-in-the-loop (SITL) cross-check is planned.
- *Solve-time tail.* The joint fleet QP reaches 330 ms at $N = 12$ in Python. Safety is preserved by the latched fallback, but the fallback rate then depends on machine load.
- *Reproducibility under wall-clock limits.* Liveness metrics vary slightly between identical-seed runs (Section VI-E).
- *Weak search.* CEM used 8 candidates, 4 generations and 2 seeds, so the denial numbers are lower bounds on the oracle family's power. Denial is, however, already saturated.
- *Small $N = 12$ cells.* With 4–14 episodes, their Wilson intervals are wide.
- *Intent proxy.* The denial attribution uses a proxy that understates filter-induced shares for acceleration-mode oracles when the filter holds an aircraft at rest, so those shares are conservative.
- *Covert-only composition.* A7 is approximated by realising the offset at the budget, and Paper IV's δ_{adv} is empirical; it was once exceeded, by $1.17\times$, on held-out data. Detection-indifferent spoofing is out of scope.
- *No real LLM yet.* RQ1 under language attack, RQ2's mediation slack and RQ4's LLM benign cost are **[pending]**.

VIII. Ethics Considerations

This paper studies how an attacker who controls an LLM agent's inputs can deny a UAV mission. The planned campaign constructs adaptive prompt injections, so the work is dual-use. We weighed benefits and harms before running anything, following the parent manuscript's and Paper IV's practice.

a) Stakeholders and harms: The stakeholders are UAV operators, people and property under flight paths, autopilot and agent-framework maintainers, and regulators. The potential harm is that injection templates or the attacker search are reused against deployed agents.

b) Why the benefit outweighs the risk:

- The attack capability is not new. Physical prompt injection against drone landing and tracking is already published [11], as are strong learned injectors [15].
- The central result *bounds* what such an attacker achieves: safety is preserved, and the practical output is a defensive configuration procedure.
- Every attack is paired with its mitigation in the same paper.
- All experiments run in simulation, and no radio-frequency energy is emitted. The pilot uses no LLM at all.
- No human subjects or personal data are involved. Any human red-team (A5) protocol will be submitted for ethics review before it runs.

c) Handling of attack material: Injection strings, the template library and attacker prompts will not be published. They are stored outside the repository, and results reference them by keyed hash only. Released oracle parameters act in a simulator's action space and confer no capability against a real language model. If the campaign finds a transferable injection against a named provider's model, we will notify the provider at least 90 days before publication.

IX. Artifact Availability

The pilot code accompanies this paper: the plant, admissible-set constructor, filter, verifier, planners, oracle search, campaign runner and statistics. Every constant carries a provenance tag (parent, Paper III, Paper IV, platform or model), all seeds are fixed, and one script regenerates every number and figure in this paper. Following programme practice, the repository is used read-only for the programme's constants. An anonymised artifact link is withheld for review. The code will be released in the programme's public repository together with an interactive *Safety Filter* page on the programme's platform. That page replays the committed results, shows the agent's proposal, the filtered action and the admissible set, and carries an "illustrative dynamics" badge. LLM completions from benign cells will be released as a replayable cache. Attack-cell completions will not (Section VIII).

X. Related Work

a) Safety filters, runtime assurance and barrier functions: Predictive safety filters minimise deviation from a desired input subject to predicted constraint satisfaction and a control-invariant terminal set [23], [24], [69], [70]. Predictive control barrier functions add recovery from infeasibility [71]. Hsu, Hu and Fisac unify Hamilton–Jacobi (HJ), control barrier function (CBF) and predictive filters as monitor plus intervention [72],

and Fisac et al. give a least-restrictive HJ framework for learning-based control [73]. CBF quadratic programs [60], [74], their input-to-state-safe variants [75], multi-robot barrier certificates [76] and MPC with discrete-time CBFs [61] are the main alternatives to our predictive QP. Robust tube MPC supplies the mismatch margin [34]–[37]. On the RTA side, Simplex and its descendants switch to a verified backup [21], [22], [38], and active-set-invariance filters and backup CBFs perform optimisation-based intervention [66], [67], [77]. Shielding plays the same role for RL agents [78], and the placement of the safeguard in the environment follows Markgraf et al. [79] and Paper III [29]. All of this work treats the protected component as imperfect, not adversarial, except where the attack targets the filter's sensors: stealthy deactivation of safety filters [32], secure safety filters for flight under sensor attack [80], and secure estimation [81]. These mark the boundary of our claim.

b) LLM robot and UAV agents, and attacks on them: LLMs act as skill selectors, code writers and action generators in robots [2]–[4], [82], and as planners or controllers for UAVs [1], [5], [6]. Attacks include jailbreaks [8]–[10] and physical indirect injection [11]. Benchmarks show that alignment does not deliver physical safety [12], and a survey lists continuous dynamics as an open gap for symbolic defences [83].

Existing guardrails use semantic or temporal-logic checks [16], [84], conformal uncertainty [85], LLM judges with LLM-parameterised CBFs [64], or dynamics-level filters: reachability [17], [18], a fuzzy CBF for UAV decisions [19], a minimal-correction filter for drone swarms [20] and a rule-based LiDAR gate [86]. Table XI in Section E positions Paper V. To our knowledge, none of the dynamics-level filters is evaluated against adaptive prompt injection, and the only adaptively evaluated guardrail is semantic. These are gaps our searches found, not exhaustively verified absences.

c) Prompt injection, defences and adaptive evaluation: Prompt injection was introduced as goal hijacking [87] and as indirect injection [7]. It has since been formalised and benchmarked [51], [88], optimised with gradients [56] and learned with RL [15]. Agent benchmarks such as AgentDojo [49], InjecAgent [50] and ASB [89] fix the metrics we adapt.

Model-level defences include structured queries [90], preference optimisation [91], spotlighting [62], the instruction hierarchy [63], detectors [92] and masked re-execution [93]. They repeatedly fall to adaptive attacks [13], [14], the latest instance of a decade-old lesson [45]–[48]. System-level defences bound what a compromised model can do: the Dual LLM pattern [94], CaMeL [26], information-flow control [27], isolation [95], privilege control [96] and design patterns [28].

Paper V is the *physical-layer* member of this second family. It inherits the family's argument that robustness should not rest on the model, and it exposes the family's residual, availability, in physical form. The residual has precedents: sponge examples [97], agent malfunction amplification [41] and classical denial of service [98]. The programme's own thread, "denial is the residual", runs from the parent's persistent-denial boundary [25] through Paper I's denial adversary [99] to Proposition 2. GPS spoofing that captures a UAV [100] is the composition case we exclude.

XI. Conclusion

We made the physical safety of a language-model-driven UAV fleet independent of the language model. A predictive safety filter whose planner input enters only its cost keeps the true state safe for every planner sequence. That holds under a stated mismatch bound, a sound and rate-bounded certificate with look-ahead, a covert sensing budget and verified execution. The proof is short; its value is the security reading and the explicit list of where it goes silent. The pilot confirms the theorem empirically against an action-space oracle that dominates every prompt-injection attacker. It also confirms each failure mode: violations return as soon as the mismatch bound, the certificate envelope, the look-ahead or the initial certification is removed.

The honest headline is a pair. Safety is unconditional under the stated model; liveness is not. A planner that merely proposes safe holds denies every mission, and no physical filter can stop it. A filter-aware attacker turns its blocked crash into filter-induced denial. Under the full sensing budget, the admissible set can be empty before take-off. What remains is the measurement that the pre-registered LLM campaign will supply: how much of the oracle's denial an attacker can actually realise through text.

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APPENDIX A
ADDITIONAL PROOFS

a) Index bookkeeping for Lemma 2: Node k of a plan created at t_j with tube age a lies at time $t_j + kT_\mathrm{s}$, at fine-step age $a + kS$. Interval k covers fine steps $a + kS + 1, \dots, a + (k{+}1)S$. The tightening $\beta_{i,k}$ in (11) uses the largest values that $\tilde{\rho}$ and ϵ_p reach on that interval, namely $\tilde{\rho}(t_j; (k{+}1)T_\mathrm{s})$ and $\epsilon_p(a + (k{+}1)S)$. This is admissible because both are non-decreasing in their time argument. The speed bound at node

$k \geq 1$ uses $\epsilon_v(a+\min\{k+1, N_\mathrm{p}\}S)$, which covers the interval that follows the node. Within an interval, the nominal velocity is affine in time, so its magnitude is bounded by the larger node value.

b) Separation along the relative chord: For a pair (i, l) on interval k, define $f(y) = n^\top y$ with $y = p_i - p_l$ and $n = n_{il,k}$. The nominal relative position is $y_z(n) = z_i(n) - z_l(n)$, and per axis it deviates from the chord between the node values $y_{z,k}$ and $y_{z,k+1}$ by at most $2b_\mathrm{ch}$. Constraint (12d) at $q = k$ and $q = k+1$ gives $f(y_{z,q}) \geq d_\mathrm{sep} + \|n\|_1(\bar{\beta}_{i,kq} + \bar{\beta}_{l,kq}) \geq d_\mathrm{sep} + \|n\|_1(\beta_{i,k} + \beta_{l,k})$. By linearity the same bound holds on the chord. Adding the errors, each axis of the true relative position deviates from the chord by at most $\beta_{i,k} + \beta_{l,k}$, since each β already contains b_ch. Hence $f(p_i - p_l) \geq d_\mathrm{sep}$. Because $\|n\|_2 = 1$, Cauchy–Schwarz gives $\|p_i - p_l\|_2 \geq f(p_i - p_l) \geq d_\mathrm{sep}$.

c) Proof of Lemma 3: Let the latched plan be certified at t_j with age a. At t_{j+1} the continuation problem starts at $z'_0 = z_1$, with age $a' = a + S$. Consider the candidate in the lemma.

- *Dynamics.* The candidate satisfies (12b), since $z'_{k+1} = z_{k+2} = A_\mathrm{s} z_{k+1} + B_\mathrm{s} v_{k+1}$ for $k < N_\mathrm{p} - 1$, and $z'_{N_\mathrm{p}} = z_{N_\mathrm{p}} = A_\mathrm{s} z_{N_\mathrm{p}} + B_\mathrm{s} \cdot 0$ because $\mathrm{vel}(z_{N_\mathrm{p}}) = 0$.
- *Tube terms.* $\epsilon_p(a' + (k+1)S) = \epsilon_p(a + (k+2)S)$, which is the old tube term at node $k+1$.
- *Certificate terms.* By Assumption A2, ρ either grows by at most $L_\rho T_\mathrm{s}$ or is re-anchored below ρ_fl, so $\max\{\rho(t_{j+1}), \rho_\mathrm{fl}\} \leq \max\{\rho(t_j), \rho_\mathrm{fl}\} + L_\rho T_\mathrm{s}$. Hence $\tilde{\rho}(t_{j+1}; (k+1)T_\mathrm{s}) \leq \tilde{\rho}(t_j; (k+2)T_\mathrm{s})$, and so $\beta'_k \leq \beta_{k+1}$ for $k < N_\mathrm{p} - 1$.
- *Last nodes.* For node $N_\mathrm{p} - 1$ and the terminal node, $\beta' \leq \bar{\rho} + b_\mathrm{tot} + \epsilon_p(\infty) + b_\mathrm{ch} = \beta_{N_\mathrm{p}}$, and the old terminal node satisfies (12c) with β_{N_p}.
- *Speed and input.* The speed rows shift identically, and the new last node has zero velocity. Inputs are a subset of the old ones plus a zero.
- *Separation.* Separation rows use the old normals shifted by one interval. The new last interval reuses $n_{N_\mathrm{p}-1}$, and both of its endpoints are the old terminal nodes, which satisfy the separation with $\beta_{N_\mathrm{p}} \geq \beta'$.

If the latched plan has already run past its horizon, the candidate is the all-zero-input loiter plan at the terminal node, which is feasible by the same terminal argument. □

d) Proof of Lemma 4(c): Let $\tilde{v}$ be feasible with $\tilde{v}_0 = u^\mathrm{des}$. Then $J^\star \leq \mu \sum_i \sum_{k \geq 1} \|\tilde{v}_{i,k}\|_2^2 \leq \mu N (N_\mathrm{p} - 1)\, 2u_\mathrm{nom}^2$. Since $\|v_0^\star - u^\mathrm{des}\|_2^2 \leq J^\star$, the claim follows. With $\mu = 10^{-4}$ the bound is loose. The empirical check (1031/1032 agreements at $\eta = 0.01\,\mathrm{m/s}^2$) shows that the regulariser almost never trades first-step fidelity for tail effort.

TABLE VII
ALL PILOT CONSTANTS WITH PROVENANCE TAGS (FROM THE PILOT'S `CONSTANTS_PAPERV.CSV`).

Name	Value	Provenance
γ (supremum)	1.6250 m/s	parent (674 samp.)
$\hat{L}_\mathrm{loc}$	1.1814	parent (local Lipschitz)
T_c	1.0 s	parent
$e^{\hat{L}_\mathrm{loc} T_\mathrm{c}}$	3.2589	derived
L_ρ	5.2958 m/s	derived (Paper III slope)
ρ_fl	0.9426 m	derived (0.178 s residual)
$\bar{\rho}$	4.9992 m	model: $L_\rho \bar{\Delta}$
$\bar{\Delta}$	0.944 s	model
δ_adv (pilot)	4.371 m	Paper IV pilot value
δ_adv (re-pooled)	5.463/5.038 m	Paper IV (FPR 1–5/10%)
$e^{\hat{L}_\mathrm{loc} T_\mathrm{c}} \delta_\mathrm{net}(0.25\,\mathrm{s})$	2.6479 m	Paper IV network tube
$s^\star(0.1)$	5.699 m	union bound, Paper IV marginals
hidden residual median	3.942 s	parent hop ledger
N_max	12	platform
T	70.0 s	platform (parent profile)
κ_sched	0.10 (0.25)	parent (platform)
δt, T_s, N_p	0.1 s, 0.5 s, 10	model
u_max	4.0 m/s^2	Paper IV
v_max	12.0 m/s	model
k_p, k_d	1.2, 1.8	Paper IV
$\bar{w}$	0.5 m/s^2	model
d_sep	5.0 m	model
goal radius, lanes, goal x	10 m, 40 m, 560 m	model
cruise speeds	10.0 / 8.5 m/s	model
solver buffers	0.10, 0.05, 0.05	model (unverified)
OSQP / total budget	40 ms / 250 ms	model
μ, η	10^{-4}, 0.01 m/s^2	model
CEM (cand., elites, gens, seeds)	8, 3, 4, 2	model

APPENDIX B
FULL PARAMETER TABLE

APPENDIX C
NOTATION

APPENDIX D
ACRONYMS

APPENDIX E
ASSURANCE AND POSITIONING TABLES

APPENDIX F
ORACLE FAMILY PER VARIANT

The CEM optima were

$$\theta_\mathrm{viol} = (1.634, -0.224, 0.563, 2.378, 0.838, -0.495),$$
$$\theta_\mathrm{den} = (1.745, -0.150, 0.113, -3.142, 0.934, 1.527),$$

with training scores -1.494 (the negative of the minimum true margin) and 1.267. The training seeds were 2000–2001 and 2100–2101, at $N = 4$.

TABLE VIII
NOTATION.

Symbol	Meaning
$\mathcal{N}$, N	fleet and its size
$\hat{x}_i = (\hat{p}_i, \hat{v}_i)$	navigation state (position, velocity)
p_i	true position (2)
e_i^{cert}, e_i^{adv}	certified estimation error; covert sensing offset
u_i, d_i, $\bar{w}$	actuator command; mismatch disturbance; its bound
δt, T_{s}, S	plant step; filter period; $S = T_{\mathrm{s}}/\delta t$
N_{p}	filter horizon (intervals); H is reserved for relay hops
$z_{i,k}$, $v_{i,k}$	nominal node state and nominal acceleration
K, Φ, λ_{c}	tube gain; error dynamics; spectral radius of Φ
ξ_i	tracking error $\hat{x}_i - z_i$
$\epsilon_p, \epsilon_v, \epsilon_u$	tube bounds (9)
$\mathcal{C} = \{h_r^\top p \le c_r\}$	keep-in polytope; d_{sep} separation
$\rho_i(t)$	protection level (Paper III)
$\Delta_i(t)$, $\bar{\Delta}$	staleness; declared staleness envelope
L_ρ, ρ_{fl}, $\bar{\rho}$	staleness slope; benign floor; cap
$\tilde{\rho}_i(t;\tau)$	look-ahead protection level (10)
γ, $\hat{L}_{\mathrm{loc}}$, T_{c}	parent's rate, Lipschitz constant, horizon
δ_{net}, δ_{adv}, $s^\star(\varepsilon)$	Paper IV network, stealth and stochastic terms
$b_{\mathrm{tot}}(\varepsilon)$, b_{adv}	scalar sensing tightening; realised offset bound
$\beta_{i,k}$, $\hat{\beta}$, $\bar{\beta}$	tightening (11) and its node/pair forms
b_{ch}	chord bulge $u_{\mathrm{nom}} T_{\mathrm{s}}^2/8$
u_{nom}, $u_{\max}$, $v_{\max}$	nominal and actuator acceleration limits; speed limit
u^{des}, π_{nom}, a_t	desired input; nominal map; planner action
P_{RA}, P_{C}	re-anchored and continuation problems
$\mathcal{A}_{\mathrm{adm}}$	admissible first inputs
μ, η	regulariser; intervention threshold
T, κ_{sched}	mission duration; schedule tolerance
S, L (traces)	safety and liveness properties
I_i, D_i	progress intent (13); goal distance
θ	detector operating point (Paper IV); oracle parameters in §IV

TABLE IX
ACRONYMS.

ADMM	alternating direction method of multipliers
ASR	attack success rate
BIR	benign intervention rate
CBF	control barrier function
CEM	cross-entropy method
CHAI	Command Hijacking against Embodied AI
CI	confidence interval
CMA-ES	covariance matrix adaptation evolution strategy
CP	Clopper–Pearson
CR	continuation rate
CRN	common random numbers
FPR	false-positive rate
FR	fallback rate
GCG	greedy coordinate gradient
GNSS	global navigation satellite system
HJ	Hamilton–Jacobi
IPM	interior-point method
IR	intervention rate
JSON	JavaScript Object Notation
LLM	large language model
LTL	linear temporal logic
MCR / MDR	mission completion / denial rate
MPC	model predictive control
OSQP	Operator Splitting Quadratic Program (solver)
PSF	predictive safety filter
QP	quadratic program
RL	reinforcement learning
RTA	runtime assurance
RTL	return to launch
SITL	software in the loop
SVR	safety-violation rate
TCB	trusted computing base
UAV	uncrewed aerial vehicle

TABLE X
DISCHARGE OF PAPER II'S L_{mon} OBLIGATIONS.

Obligation	How Paper V discharges it	What remains open
O_{mcov} coverage	proof (Theorem 1); the monitor is model-based, not learned	rests on certificate soundness (Assumption A2) and the calibrated γ (transfer)
O_{lat} latency	measured: p99 80.3/206.8 ms vs 500 ms; overrun $\rightarrow$ latched tail	compiled solver for high rates
O_{rec} recovery	proof: robustly invariant loiter set plus latched tail	RTL is certified only if filtered
O_{ind} independence	architectural: the filter never reads LLM text	fails under non-covert sensing attacks (A7 beyond Paper IV's class)

TABLE XI
POSITIONING AGAINST LLM-ROBOT GUARDRAILS.

Work	Layer	LLM-indep.	Adaptive eval.	UAV
RoboGuard [16]	LTL + root-of-trust LLM	partial	yes	no
Hafez et al. [17]	reachability	yes	no	no
SanDRA [18]	reachability + rules	yes	no	no
SAFER [64]	LLM judge + CBF	no	no	no
SAGE-LLM [19]	fuzzy CBF	claimed	not reported	yes
SwarmGPT [20]	min.-correction filter	yes	no	yes
Secure SF [80]	robust CBF, sensor attack	n/a	sensor only	yes
This paper	predictive filter	**yes**	**yes** (protocol)	**yes**

TABLE XII
ORACLE VARIANTS WITH THE FILTER ON (16 EPISODES PER HAND-DESIGNED CELL FOR $N = 4$, 5 FOR $N = 12$; 40/14 FOR THE CEM OPTIMA). MARGINS ARE THE CLOSEST TRUE APPROACHES IN METRES.

N	Variant	Corr.	Sep.	FR	Denial	filter
4	oracle_denial (CEM)	0.06	6.30	0.020	1.00	0.95
4	goal+neighbour	0.72	0.76	0.001	0.56	0.56
4	goal+wall	0.13	6.61	0.002	1.00	0.50
4	overspeed	0.09	4.93	0.011	1.00	1.00
4	press_neighbour	14.37	0.44	0.000	1.00	0.00
4	press_wall	0.21	10.64	0.000	1.00	0.00
4	oracle_violation (CEM)	0.10	9.87	0.008	1.00	0.00
12	oracle_denial (CEM)	0.06	4.85	0.124	1.00	1.00
12	goal+neighbour	0.19	0.70	0.061	1.00	1.00
12	goal+wall	0.08	1.23	0.017	1.00	0.00
12	overspeed	0.12	5.08	0.094	1.00	1.00
12	press_neighbour	0.19	0.38	0.018	1.00	0.00
12	press_wall	0.17	10.54	0.002	1.00	0.00
12	oracle_violation (CEM)	0.06	1.37	0.144	1.00	0.00